Chapter 27

Conventional Mechanical Ventilation

Medicine is a science which has been more laboured than advanced.
For I find much iteration, but small addition.

Sir Francis Bacon (*a*)

The earliest ventilators (introduced in the late 1920s) were negative pressure chambers known as *iron lungs* that enclosed the patient from the neck to the ankles. These behemoths (early models weighed about a ton) were the sole means of ventilatory support until the polio epidemic in the 1950's, when the demand for assisted ventilation exceeded the supply of iron lungs. In Copenhagen, Denmark, tracheostomies were introduced, and medical students worked in 8 hour shifts as human ventilators, manually inflating the lungs of afflicted patients (1). In Boston, the local Emerson Company had a prototype device for positive pressure lung inflations, which was put to use at the Massachusetts General Hospital, and became an instant success.

Fast forward about 70 years, and there are estimated 174 different modes of ventilation (2). However, only one has shown evidence of improved clinical outcomes (3,4), and only because it limits the lung damage produced by positive pressure ventilation (see Chapter 24). What this means is that mechanical ventilation is much more complicated than it needs to be, and that "less is better" with this form of support.

This chapter describes the basic methods of positive pressure ventilation. Although far fewer than the bloated number of available methods, they will provide effective ventilatory support for most (if not all) of your patients.

THE VENTILATOR BREATH

There are two basic methods of positive-pressure lung inflation. These are summarized below, and are illustrated in Figure 27.1.

- *Volume-control ventilation*, where the desired inflation volume (tidal volume) is preselected. The gas is delivered at a constant flow rate, and there is a steady rise in volume and airway pressure until the target volume is reached. Exhalation is then passive.
- *Pressure-control ventilation*, where the desired inflation pressure is preselected. In this case,

the inspiratory flow rate has a decelerating pattern, with the peak flow rate in the initial phase of inspiration, and the flow rate then decreasing to zero by the end of inspiration. Because of this flow profile, the increments in lung volume and airway pressure are greater in the early phase of inspiration. The desired (peak) pressure is reached early, and a preselected inspiratory time then determines the end of inspiration. Exhalation occurs passively.

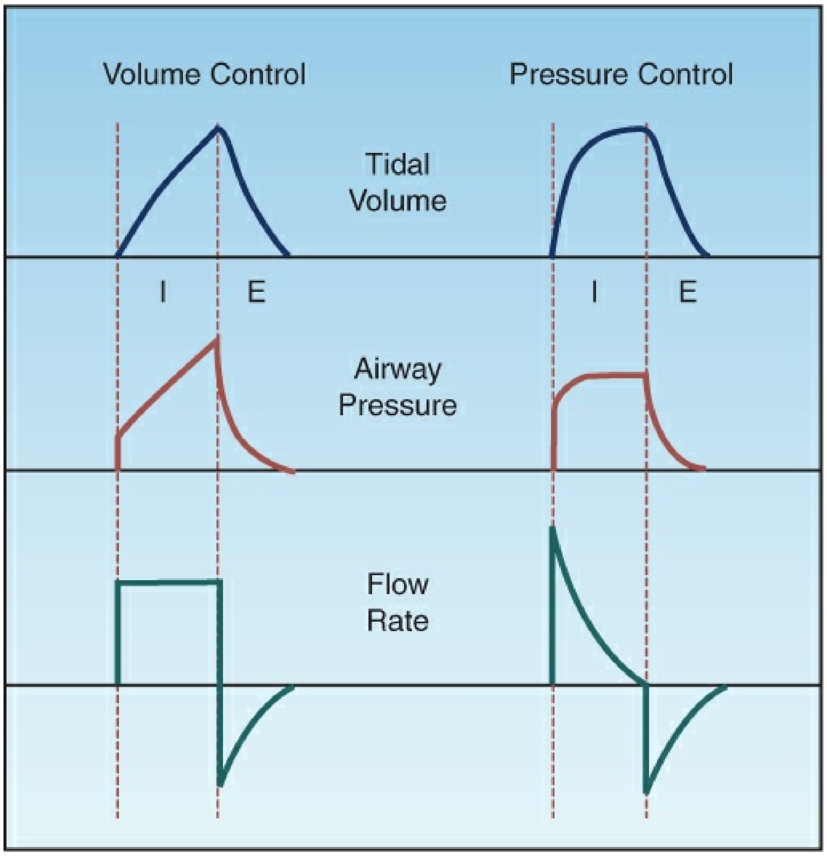

FIGURE 27.1 Changes in lung volume, airway pressure, and flow rate during a single ventilator breath using volume control and pressure control methods of lung inflation. I = inspiration, E = expiration. See text for further explanation.

All variables (i.e., volume, pressure, and flow) are recorded and displayed by the ventilator, but the measurements are made at the level of the endotracheal tube, and may not reflect conditions in the distal airspaces (the alveoli). In addition, the airway pressures are measured as *transthoracic* pressures (i.e., airway pressure relative to atmospheric pressure), and are influenced by both the lungs and the chest wall. The measurement of *transpulmonary* pressure (i.e., airway pressure relative to pleural pressure) eliminates the influence of the chest wall, but requires an esophageal balloon to measure the intrapleural pressure.

Volume Control

Volume-control ventilation (VCV) delivers a preselected tidal volume at a constant flow rate, and the peak airway pressure (Ppeak), is determined by the mechanical properties (resistance and elastance) of the lungs and chest wall. This can be expressed as follows:

$$\text{Ppeak} = \text{Pres} + \text{Pel} \tag{27.1}$$

where Pres is the pressure attributed to airflow resistance in the airways, and Pel is the pressure attributed to the elastic recoil force of the lungs and chest wall (elastance). (*Note:* Elastance is the resistance of an object to being deformed.)

The Plateau Pressure

The relative contributions of resistance and elastance to the peak airway pressure can be identified by briefly occluding the ventilator circuit at the end of lung inflation (5). This "inflation-hold" maneuver is illustrated in Figure 27.2. The peak pressure decreases initially during the inflation hold, and then reaches a constant *plateau pressure* until the occlusion is released and exhalation proceeds. Since there is no airflow during the inflation hold maneuver, the plateau pressure is equivalent to the pressure in the alveoli (Palv), and this is the pressure generated by the elastic recoil force of the lungs and chest wall (Pel): i.e.,

$$\text{Pplateau} = \text{Palv} = \text{Pel} \tag{27.2}$$

(*Note:* Positive end-expiratory pressure or PEEP is used commonly during mechanical ventilation, as described later, and this must be subtracted from the plateau pressure to obtain the actual alveolar pressure.) The difference between the peak and plateau pressures is then the pressure needed to overcome the resistance to airflow (Pres): i.e.,

$$(\text{Ppeak} - \text{Pplateau}) = \text{Pres} \tag{27.3}$$

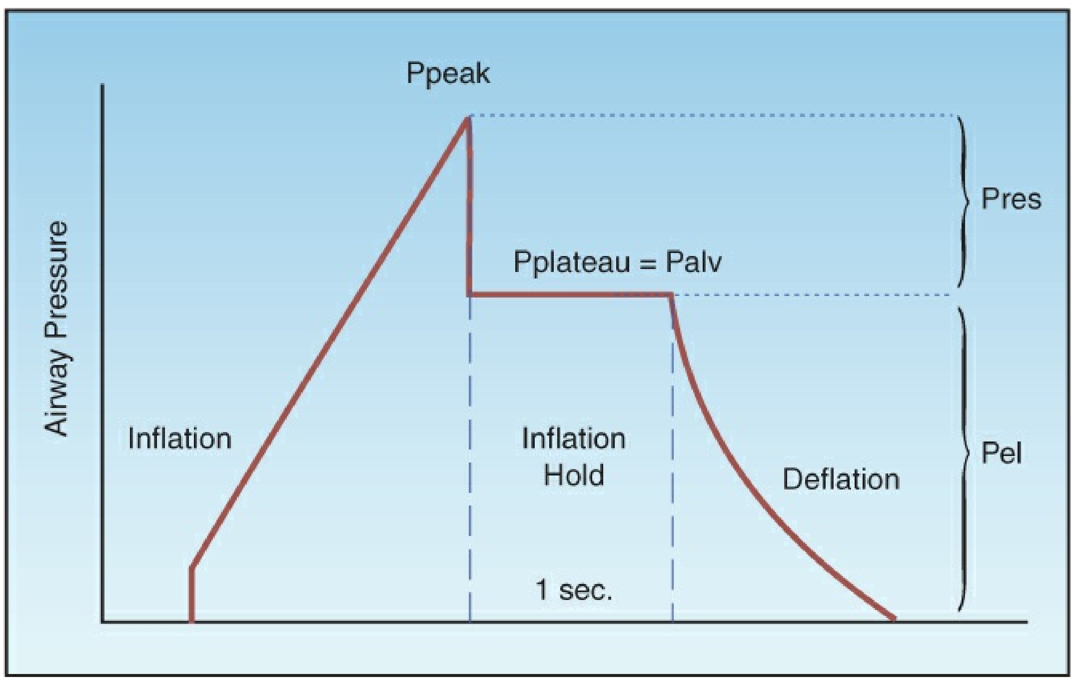

FIGURE 27.2 Airway pressure profile for a volume-controlled lung inflation with a brief end-inspiratory occlusion (inflation-hold). Ppeak is the peak airway pressure, Pplateau is the end-inspiratory occlusion pressure, Palv is the alveolar pressure, Pres is the pressure attributed to airway resistance, and Pel is the pressure attributed to the elastic recoil force of the lungs and chest wall. See text for explanation.

RISK OF ALVEOLAR INJURY: The peak airway pressure is higher with VCV than with pressure control ventilation (see Figure 27.1), and there is a misconception that this increases the risk of alveolar rupture. However, this risk is a function of the alveolar pressure; i.e., the plateau pressure. Clinical studies indicate that *the risk of alveolar rupture is negligible if the alveolar pressure (plateau pressure) is* ≤ 30 *cm* H_2O (2,4), and this is one of the major goals of *lung protective ventilation* (2), which is described later (see Table 27.1).

Advantages

The major advantage of VCV is the ability to maintain a constant tidal volume despite changes in the mechanical properties of the lungs. This is especially important for limiting the tidal volumes used during mechanical ventilation, which is the basis of lung protective ventilation.

Disadvantages

The constant inspiratory flow rate during VCV creates two potential disadvantages. First, the duration of inspiration is relatively short, and this can lead to uneven alveolar filling. Second, the maximum inspiratory flow is limited when flow is constant; as a result the inspiratory flow rate can be inadequate for patients with high ventilatory demands. A decelerating flow pattern is available for VCV, and has been shown to improve patient comfort (6).

Pressure Control

With pressure control ventilation (PCV), the desired inflation pressure is preselected, and a decelerating inspiratory flow rate provides high flows at the onset of the lung inflation, to attain the desired inflation pressure quickly. The inspiratory time is adjusted to allow enough time for the inspiratory flow rate to fall to zero at the end of inspiration. *Since there is no airflow at the end of the inspiration, the end-inspiratory airway pressure is equivalent to the alveolar pressure.*

Advantages

The major advantage of PCV is the increased patient acceptance and reduced risk of ventilatory asynchrony when compared to VCV (7). This advantage is attributed to the high initial flow rates and the longer duration of inspiration with PCV.

Disadvantages

The major disadvantage with PCV is the change in tidal volume that occurs with changes in airway resistance or lung elastance, since this can increase the risk of alveolar overdistension and ventilator-induced lung injury (see Chapter 24). However, this disadvantage is corrected with the adaptive mode of ventilation described next.

Pressure-Regulated, Volume Control

Pressure-regulated, volume control ventilation (PRVC) is an adaptive mode of ventilation that provides a constant tidal volume (like volume control) but limits the end-inspiratory airway pressures (like pressure control). PRVC operates like an intelligent form of volume control; i.e., the ventilator monitors lung compliance, and uses these measurements to select the lowest airway pressure needed to deliver the desired tidal volume (8). This has become a popular mode of mechanical ventilation, but patients with a high ventilatory drive may not receive the needed inspiratory flow rate, leading to ventilator asynchrony (9).

End-Expiratory Pressure

The end-expiratory pressure is the minimum pressure in the alveoli during a ventilatory cycle. The different forms of end-expiratory pressure are illustrated in Figure 27.3.

ZEEP

During appropriate ventilation in the normal lung, there is no airflow at the end of expiration, and the pressure in the alveoli is equivalent to atmospheric pressure. Since atmospheric pressure is a zero reference point for breathing, this condition is called *zero end-expiratory pressure*, or ZEEP.

Applied PEEP

Positive end-expiratory pressure (PEEP) can be added through the ventilatory circuit (via a pressure-sensitive valve in the expiratory limb of the circuit) so that exhalation will cease when the airway pressure falls to the preselected PEEP level. Applied PEEP is used routinely during mechanical ventilation to prevent the collapse of small airways at the end of expiration, and to open collapsed alveoli (recruitment). The uses, advantages, and disadvantages of applied PEEP are described later in the chapter.

Occult PEEP

When there is continued airflow at the end of expiration, the lungs do not completely empty, and the alveolar pressure remains positive even though the proximal airway pressure falls to atmospheric (zero) pressure. This pressure is sometimes called intrinsic PEEP or auto-PEEP, but *occult PEEP* is also used because the PEEP *is not apparent in proximal airway pressure recordings* (10). Occult PEEP can be the result of dynamic hyperinflation in patients with asthma and COPD, as described in Chapter 23, or it can be the result of ventilator settings that predispose to end-expiratory airflow (e.g., high tidal volumes, decreased time for exhalation).

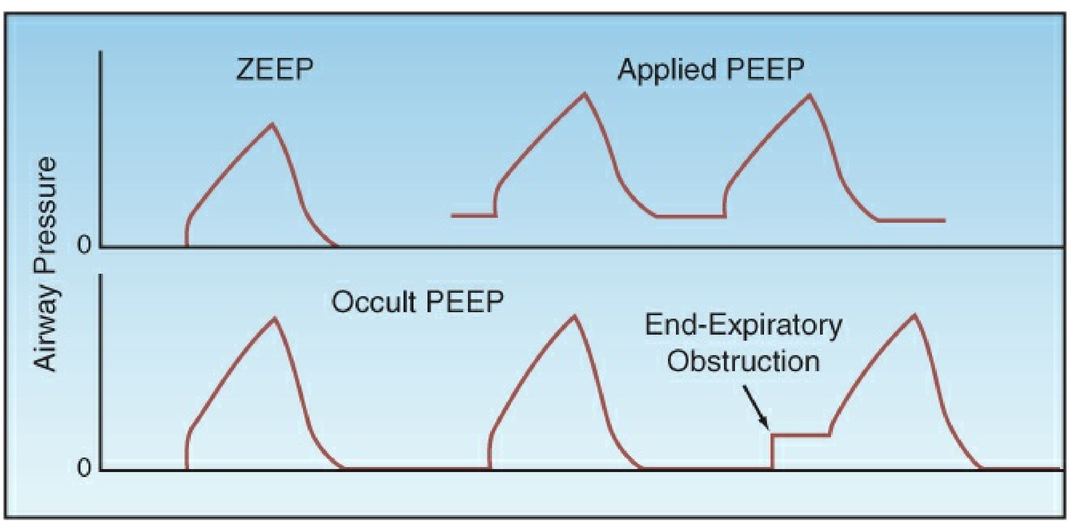

FIGURE 27.3 Different types of end-expiratory pressure. See text for explanation.

Occult PEEP can be detected on the flow tracing by noting the presence of airflow at the end of expiration, as illustrated in Figure 23.5. If occult PEEP is present, it can be quantified by occluding the expiratory circuit at the end of expiration. During an end-expiratory occlusion, alveolar pressure will equilibrate with proximal airway pressure, and the occult PEEP becomes apparent as a abrupt increase in airway pressure, as shown in Figure 27.3. This maneuver is successful *only in patients who have no respiratory efforts*. Occult PEEP is described in more detail in Chapter 28.

Mean Airway Pressure

The mean airway pressure is the average pressure in the airway during the ventilatory cycle, and is influenced by several variables, including the peak airway pressure, the contour of the pressure waveform, the PEEP level, the respiratory rate, and the inflation time relative to the total time of the ventilatory cycle (T_I /T_{tot}). The mean airway pressure that is displayed by ventilators is obtained by integrating the area under the airway pressure waveform.

The mean airway pressure is linked to the hemodynamic effects of positive pressure ventilation. (The intrapleural pressure is the important influence on cardiac function, but this pressure requires measurement with an intraesophageal balloon, and is not monitored routinely.) Typical values for mean airway pressure during positive pressure ventilation are 5–10 cm H_2O for normal lungs, 10–20 cm H_2O for airflow obstruction, and 20–30 cm H_2O for noncompliant (stiff) lungs (11).

Thoracic Compliance

Compliance (Δvolume/Δpressure) is the reciprocal of elastance, and is the traditional term used to express the elastic properties of structures with chambers (like the heart and lungs). Compliance expresses *distensibility*, or the tendency of a chamber to increase in volume when exposed to a given distending pressure. The compliance that is measured during mechanical ventilation is a *thoracic compliance*, and includes both the lungs and the chest wall.

Volume-Control Ventilation

During VCV, the static compliance of the thorax (Cstat) is expressed as the preselected tidal volume (VT) divided by the difference between the plateau pressure and the total PEEP level (applied plus occult PEEP):

$$\text{Cstat} = V_T/[\text{Pplateau} - \text{PEEP(tot)}] \tag{27.4}$$

This is a “static” compliance because the pressures involved are measured in the absence of airflow. In patients with normal lungs, Cstat is 50–80 mL/cm H_2O (12), and in patients with infiltrative lung diseases (e.g., pulmonary edema or acute respiratory distress syndrome), Cstat is typically <25 mL/cm H_2O (13).

Pressure-Control Ventilation

During PCV, Cstat is the exhaled tidal volume (exhaled V_T) divided by the difference between the end-inspiratory airway pressure (Paw) and the total PEEP level:

$$\text{Cstat} = \text{Exhaled } V_T/[\text{Paw(end-insp)} - \text{PEEP(tot)}] \quad (27.5)$$

Sources of Error

- During passive ventilation, the chest wall can account for 35% of the total thoracic compliance (14,15), and this contribution increases when the chest wall muscles contract. Therefore, to avoid errors in interpreting changes in thoracic compliance, the *compliance measurements should be performed in patients with minimal or no breathing efforts*.
- Thoracic compliance is volume-dependent; i.e., it decreases as lung volume increases. Absolute lung volumes cannot be measured during mechanical ventilation, so *compliance measurements should be performed at the same tidal volume*.
- The tidal volume used for the compliance measurement should be adjusted for the compliance of the ventilator tubing, which is typically 3 mL/cm H_2O (13). For example, if the preselected tidal volume during VCV is 500 mL and the peak airway pressure is 40 cm H_2O, then 3 × 40 = 120 mL of the delivered volume will be lost to expansion of the ventilator tubing, and the actual tidal volume reaching the patient will be 500 – 120 = 380 mL. When using exhaled tidal volumes during PCV, the airway pressure at end-inspiration should be used for the volume adjustments.

ASSIST-CONTROL VENTILATION

Assist-control ventilation (ACV) allows the patient to initiate the ventilator breath, but if this is not possible, ventilator breaths are delivered at a preselected rate (controlled or time-triggered ventilation). The ventilator breaths during ACV can be volume-cycled or pressure-cycled.

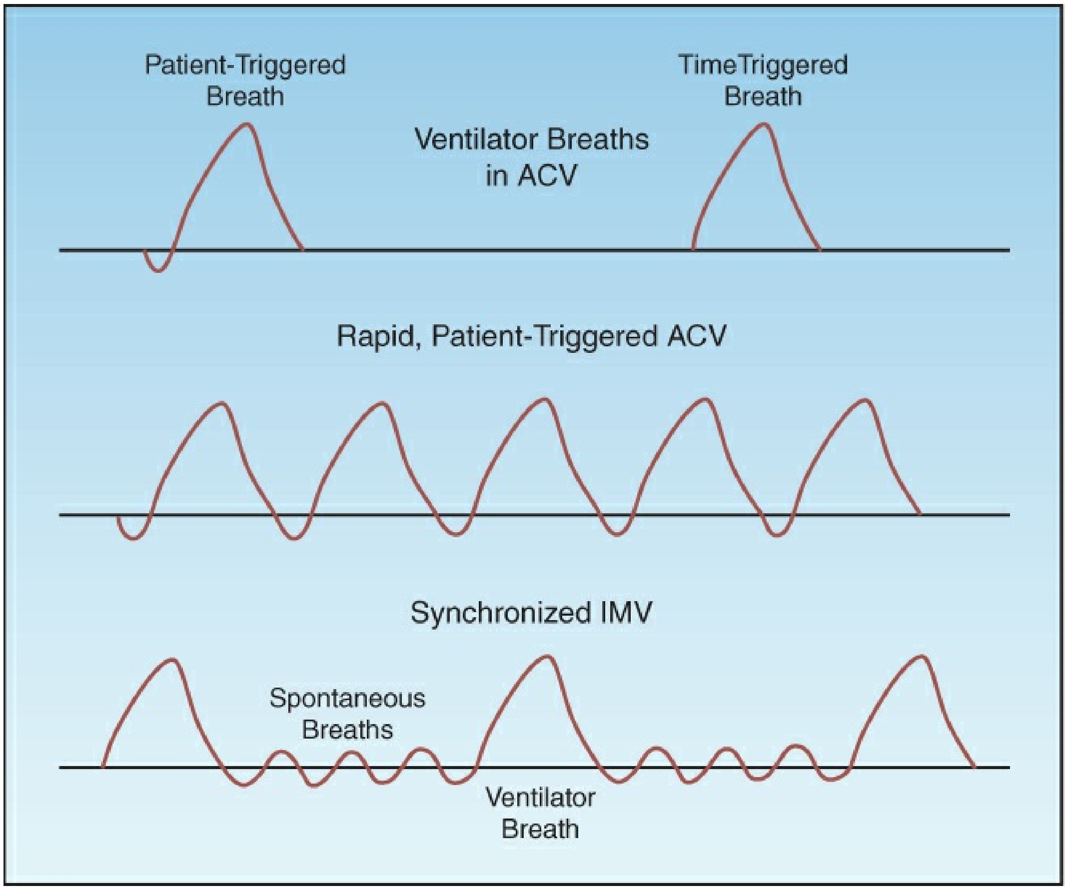

FIGURE 27.4 Airway pressure patterns in assist-control ventilation (ACV) and synchronized intermittent mandatory ventilation (SIMV). See text for further explanation.

Triggers

The ventilator breaths in ACV can be initiated (triggered) in two ways, as shown in the upper panel of Figure 27.4. The pressure waveform on the left is preceded by a negative pressure deflection, which represents a spontaneous inspiratory effort by the patient. This is a *patient-triggered* ventilator breath. The pressure waveform on the right is not preceded by a negative pressure deflection, indicating the absence of a spontaneous inspiratory effort. In this case, there is no interaction between the patient and the ventilator, and the ventilator breaths are delivered at a preselected rate. This is a *time-triggered* ventilator breath.

Patient-Related Triggers

There are two types of signals that allow patients to trigger a ventilator breath: negative pressure and inspiratory flow rate.

NEGATIVE PRESSURE: Patients can trigger a ventilator breath by generating a negative airway pressure of 2 to 3 cm H_2O, which opens a pressure-sensitive valve in the ventilator. Despite this low pressure requirement, about one-third of inspiratory efforts fail to trigger a ventilator breath when negative pressure is the trigger signal (16).

FLOW RATE: Flow triggering involves little or no change in pressures and volumes, and thus involves less mechanical work than pressure triggering. For this reason, flow has replaced pressure as the standard trigger mechanism. The flow rate that is required to trigger a ventilator breath differs for each brand of ventilator, but rates of 1–10 L/min are usually required. Auto-triggering from system leaks (which create flow changes) is the major problem associated with flow triggering (17).

The Respiratory Cycle

A general rule of thumb during mechanical ventilation is to allow at least twice the amount of time for expiration as allowed for inspiration. This is equivalent to an inspiration:expiration time ratio (I:E ratio) of at least 1:2. The goal is to allow enough time for complete exhalation to prevent dynamic hyperinflation and intrinsic or occult PEEP (see Figure 23.5) If the duration of exhalation is too short, the I:E ratio can be increased by: (a) increasing the inspiratory flow rate, (b) reducing the tidal volume, or (c) decreasing the inspiratory time (for pressure control).

Rapid Breathing

When each breath is a patient-triggered ventilator breath, rapid breathing like that shown in Figure 27.4 can severely curtail the time for exhalation and increase the risk of dynamic hyperinflation and occult PEEP. When rapid breathing is the result of a condition other than discomfort or anxiety, attempts to reduce the respiratory rate with sedation or inspiratory flow adjustments are often unsuccessful. In this situation, the ventilator mode described next may be the answer.

INTERMITTENT MANDATORY VENTILATION

Difficulties with rapid breathing during ACV in neonates with respiratory distress syndrome, who typically have respiratory rates above 40 breaths/minute, led to the introduction of *intermittent mandatory ventilation* (IMV).

The Method

IMV is designed to allow spontaneous breathing between ventilator breaths. This is accomplished by placing a spontaneous breathing circuit in parallel with the ventilator circuit, with a unidirectional valve that opens the spontaneous breathing circuit when a ventilator breath is not being delivered. The ventilatory pattern in IMV is illustrated in the lower panel of Figure 27.4. Note that the ventilator breath is delivered in synchrony with the spontaneous breath, which is called *synchronized IMV* (SIMV).

The ventilator breaths during SIMV can be volume-cycled or pressure-cycled. The rate of ventilator breaths can be started at 10 breaths/min, and then adjusted as needed to achieve the desired minute ventilation (spontaneous plus assisted breaths).

Adverse Effects

The principal adverse effects of IMV are: (a) an increased work of breathing, and (b) a decrease in cardiac output, primarily in patients with left ventricular dysfunction. Both effects are the result of the spontaneous breathing period.

Work of Breathing

The heightened work of breathing during the spontaneous breathing period in IMV is attributed to resistance in the ventilator circuit. Spontaneous breathing with pressure support, which is described in Chapter 26 (see Figure 26.2), overcomes the added resistance of the ventilator circuit and reduces the work of breathing (18). As a result, pressure-support ventilation (at 10 cm H_2O) is now used routinely during the spontaneous breathing periods in IMV.

Cardiac Output

As described in Chapter 26, positive pressure ventilation reduces left ventricular afterload and can increase cardiac output (see Figure 26.6) (19). IMV has the opposite effect and increases left ventricular afterload (during the spontaneous breathing periods), which results in a decrease in cardiac output in patients with left ventricular dysfunction (20).

Summary

The major indication for IMV is rapid breathing with incomplete exhalation during assist-control ventilation. The spontaneous breathing periods during IMV promote alveolar emptying and reduce the risk of air trapping and intrinsic PEEP. IMV can increase the work of breathing and impair cardiac output in patients with left ventricular dysfunction; as a result, *IMV is not advised for patients with respiratory muscle weakness or left heart failure*.

POSITIVE END-EXPIRATORY PRESSURE

Ventilator-dependent patients usually have an increase in airway resistance (e.g., from COPD) or a decrease in lung compliance (e.g., from lung consolidation), and both of these conditions promote the collapse of small airways and alveoli at the end of expiration. This impairs gas exchange in the lungs and promotes hypoxemia. In addition, collapsed airspaces can reopen in the ensuing lung inflation, and the repetitive opening and closing of small airways can damage the airway epithelium by generating excessive shear forces (21). This type of injury is called *atelectrauma* (22), and it is one of the mechanisms for *ventilator-induced lung injury* (which is described in detail in Chapter 24).

To combat this tendency for alveolar collapse, positive end-expiratory pressure (PEEP) is used routinely during mechanical ventilation. The standard practice is to begin with a pressure of 5–8 cm H_2O, but the optimal level of PEEP may differ for each patient, as described next.

Best PEEP

Relatively small levels of PEEP (5–10 cm H_2O) can prevent the collapse of distal airspaces, while higher PEEP levels (10–20 cm H_2O) can open airspaces that are collapsed. This latter effect is known as *alveolar recruitment*, and it increases the available surface area in the lungs for gas exchange (23). However, higher levels of PEEP can also overdistend and rupture alveoli in normal lung regions. This is known as *volutrauma*, and is the principal form of ventilator-induced lung injury (23). The following are some ways to determine whether PEEP is promoting alveolar recruitment (favorable response) or promoting alveolar overdistension (unfavorable response).

Lung Compliance

When PEEP is promoting alveolar recruitment, the compliance (distensibility) of the lungs will increase, but when PEEP is overdistending alveoli, lung compliance will decrease. Therefore, monitoring the changes in thoracic compliance (as in Equations 27.4 and 27.5) in response to incremental levels of PEEP can identify whether PEEP is beneficial or harmful. This is demonstrated in Figure 27.5 (24). The upper curve in this figure shows that compliance initially increases in response to PEEP, but eventually begins to decrease. The transition point then identifies the "best PEEP" level.

Driving Pressure

At any given tidal volume, lung compliance is a function of the difference between the peak alveolar pressure and the PEEP level (as shown in Equations 27.4 and 27.5). This pressure difference is the "driving pressure" for alveolar distension, and the changes in this pressure can identify if PEEP is promoting alveolar recruitment or overdistension. This is demonstrated in the lower curve in Figure 27.5 (24). The driving pressure decreases with lower levels of PEEP (indicating that alveolar recruitment is the predominant response), but begins to increase at higher levels of PEEP (indicating that alveolar overdistension is a risk), and the transition point is then the "best PEEP" level.

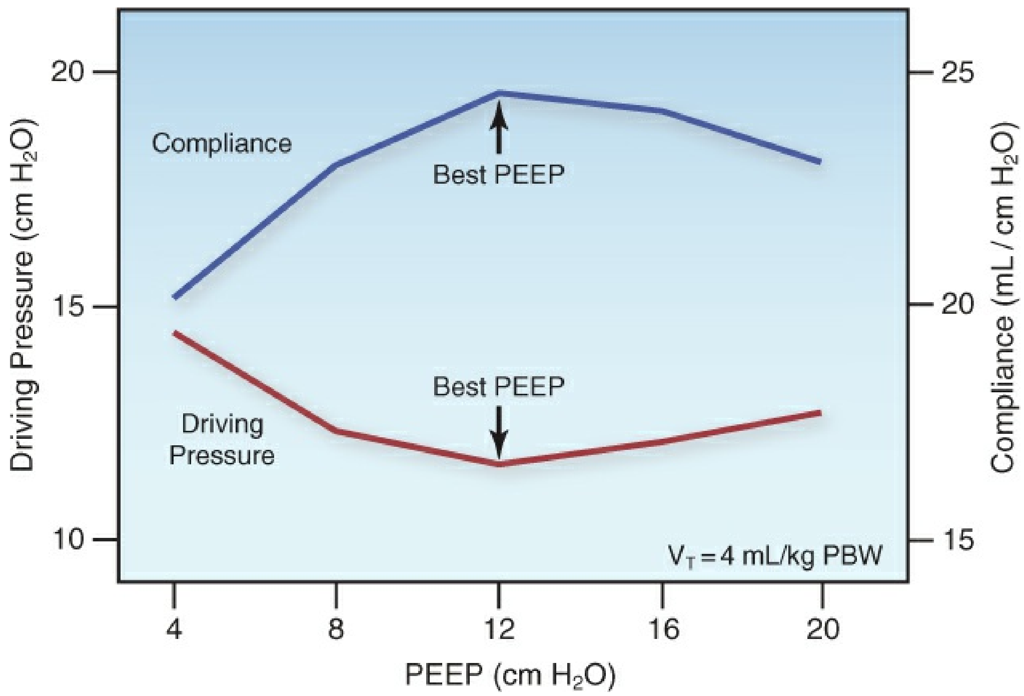

FIGURE 27.5 The effects of incremental levels of PEEP on thoracic compliance and driving pressure, which were measured at a constant tidal volume (V_T) of 4 mL/kg predicted body weight (PBW). The point of transition in each curve identifies the "best PEEP". Data from Reference 24. See text for further explanation.

PaO_2 /F_IO_2 Ratio

The relationship between the arterial PO_2 and the fractional concentration of inspired oxygen, as expressed by the PaO_2/F_IO_2 ratio, is a measure of the efficiency of gas exchange in the lungs.

When PEEP promotes alveolar recruitment, the PaO_2/FIO_2 ratio will increase, and when PEEP does not promote alveolar recruitment, the PaO_2/FIO_2 ratio will remain unchanged or will decrease. An example of a beneficial response in the PaO_2/FIO_2 ratio (indicating alveolar recruitment) is shown in Figure 27.6. However, an increase in the PaO_2/FIO_2 ratio is not synonymous with an increases in O_2 delivery to tissues, as explained next.

OXYGEN DELIVERY: The beneficial effects of PEEP on arterial oxygenation may not be accompanied by a similar benefit in oxygen delivery to tissues, because the rate of O_2 delivery in arterial blood (DO_2) is dependent on the cardiac output (CO) as well as the arterial O_2 content (CaO_2): i.e.,

$$DO_2 = CO \times CaO_2 \tag{27.6}$$

The effects of positive pressure ventilation that impair cardiac filling are described in Chapter 26 (see Figure 26.7), and PEEP magnifies these effects (25). This means that PEEP can decrease the cardiac output and negate the beneficial effects on arterial oxygenation. This is shown in Figure 27.6, where the progressive increase in PaO_2/FIO_2 ratio in response to incremental PEEP is accompanied by a progressive decrease in cardiac output (26). Thus, PEEP can promote alveolar recruitment and increase arterial oxygenation, but systemic O_2 delivery may not improve if PEEP also decreases the cardiac output.

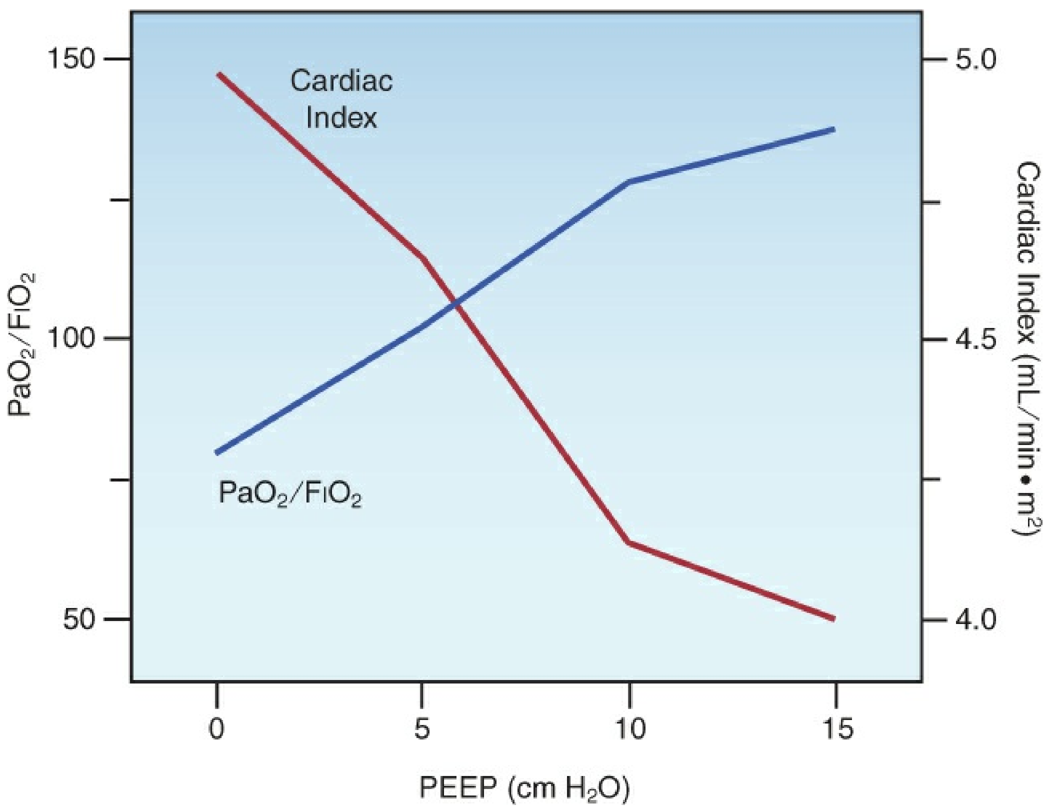

FIGURE 27.6 The opposing effects of positive end expiratory pressure (PEEP) on oxygenation (PaO_2 /FIO_2) and cardiac index in patients with ARDS. Data from Reference 26.

The opposing effects of PEEP on arterial oxygenation and cardiac output emphasize the need to include a measure of systemic O_2 delivery in the determination of "best PEEP". Since cardiac output is not frequently monitored during mechanical ventilation, the central venous O_2 saturation can be used as a surrogate measure of changes in cardiac output and O_2 delivery, as explained in Chapter 9.

VENTILATOR SETTINGS

When mechanical ventilation is initiated, the respiratory therapist will ask you for the following parameters: (a) mode of ventilation, (b) tidal volume, (c) respiratory rate, (d) PEEP level, and (e) inspired O_2 concentration. The following is a list of suggestions for setting up mechanical ventilation.

Assist-Control Ventilation

- Select assist-control as the initial mode of ventilation.
- It may be necessary to switch to synchronized IMV in patients who are breathing too rapidly in the assist-control mode (see later).

Volume vs. Pressure Control

- Use volume control of some form (either standard volume control or pressure-regulated volume control) in order to use *lung protective ventilation*, where control of the tidal volume is a necessity.
- If lung protective ventilation is not improving gas exchange in patients with ARDS, consider switching to "airway pressure release ventilation" (see Figure 24.7).

Tidal Volume

The following recommendations are from the protocol for lung protective ventilation, which is summarized in Table 27.1.

- Select an initial tidal volume of 8 mL/kg using *predicted body weight*. (The formulas for predicted body weight are in Table 27.1.)
- Reduce the tidal volume to 6 mL/kg over the next 2 hours, if possible.
- Monitor the end-inspiratory alveolar pressure (i.e., the plateau pressure) and keep it ≤30 cm H_2O (to limit the risk of volutrauma).

TABLE 27.1	Protocol for Lung Protective Ventilation
First Stage	1. Calculate patient's predicted body weight (PBW). Males: PBW = 50 + [2.3 x (height in inches – 60)] Females: PBW = 45.5 + [2.3 x (height in inches – 60)]

	2. Set initial tidal volume (V_T) at 8 mL/kg PBW. 3. Add positive end-expiratory pressure (PEEP) of 5 cm H_2O. 4. Select the lowest FIO_2 that achieves an SpO_2 of 88–95%. 5. Reduce V_T by 1 mL/kg every 2 hrs until V_T = 6 mL/kg.
Second Stage	1. When V_T = 6 mL/kg, measure plateau pressure (Ppl). 2. If Ppl >30 cm H_2O decrease V_T in 1 mL/kg increments until Ppl <30 cm H_2O or V_T. = 4 mL/kg.
Third Stage	1. Monitor blood gases for respiratory acidosis. 2. If pH = 7.15–7.30, increase respiratory rate (RR) until pH >7.30 or RR = 35 bpm. 3. If pH <7.15, increase RR to 35 bpm. If pH is still <7.15, increase V_T in 1 mL/kg increments until pH >7.15.
Optimal Goals	V_T = 6 mL/kg, Ppl ≤30 cm H_2O, SpO_2 = 88–95%, pH = 7.30–7.45.

Adapted from the protocol developed by the ARDS Network, available at www.ardsnet.org.

Inspiratory Flow Rate

- Select an inspiratory flow rate of 60 L/min if the patient is breathing quietly or has no spontaneous respirations.
- Use higher inspiratory flow rates (e.g., ≥80 L/min) for patients with respiratory distress or a high minute ventilation (≥10 L/min).

I:E Ratio

- The I:E ratio should be ≥1:2.
- If the I:E ratio is <1:2, the options for increasing the I:E ratio include: (a) increasing the inspiratory flow rate, (b) decreasing the tidal volume, or (c) decreasing the respiratory rate, if possible.

Respiratory Rate

- If the patient has no spontaneous respirations, set the respiratory rate to achieve your estimate of the patient's minute ventilation just prior to intubation, but do not exceed 35 breaths/minute.
- If the patient is triggering each ventilator breath, set the machine rate just below the patient's spontaneous respiratory rate.
- After 30 minutes, check the arterial PCO_2, and adjust the respiratory rate, if necessary, to achieve the desired PCO_2.
- For patients who are breathing rapidly and have an acute respiratory alkalosis or evidence of occult PEEP, consider switching to synchronized IMV (SIMV) as the mode of ventilation.

PEEP

. Set the initial PEEP to 5–8 cm H_2O.

. Further increases in PEEP may be necessary for the following reasons:

a. To increase the PaO_2/FIO_2 ratio in cases where a toxic level of inhaled oxygen (>60%) is required to maintain adequate oxygenation (SaO_2 ≥90%).

b. To determine the “best PEEP” for that patient.

Occult PEEP

. Check the flow rate at the end of expiration: if the expiratory flow rate has not returned to zero at end-expiration, this is evidence of dynamic hyperinflation and occult PEEP (see Figure 23.5).

. If there is evidence of occult PEEP, attempt to prolong the time for expiration by increasing the I:E ratio using the maneuvers described previously.

. If increasing the I:E ratio is not possible or is not successful, then measure the level of occult PEEP with the end-inspiratory occlusion method, and add extrinsic PEEP at a level that is just below the occult PEEP. (The rationale for this maneuver is explained in Chapter 28.)

A FINAL WORD

A Loss of Focus

Since the advent of mechanical ventilation, an enormous amount of time and energy has been devoted to the development of newer methods of ventilation (174 of them at last count), with only one of these showing evidence of improved outcomes (and only because it involves a decrease in ventilatory support) (3). What has been lost in these endeavors is the simple fact that mechanical ventilation is a temporary support measure for acute respiratory failure, and is not a treatment modality for lung disease. Therefore, if we want to improve outcomes in ventilator-dependent patients, less attention should be given to the knobs on ventilators, and more attention should be directed at the diseases that create ventilator dependency.

References

a. Bacon F. *The Advancement of Learning*, 1605.

1. Lassen HCA. A preliminary report on the 1952 epidemic of poliomyelitis in Copenhagen with special reference to the treatment of acute respiratory insufficiency. Lancet 1953; 1:37–41.
2. Cairo JM, Pilbean SP. Mosby’s Respiratory Care Equipment. 8th ed. St. Louis: Mosby Elsevier; 2010.
3. The Acute Respiratory Distress Syndrome Network. Ventilation with lower tidal volumes as compared with traditional tidal volumes for acute lung injury and the acute respiratory distress syndrome. N Engl J Med 2000; 342(18):1301–1308.
4. Petrucci N, De Feo C. Lung protective ventilation strategy for the acute respiratory distress syndrome. Cochrane Database Syst Rev 2013:CD003844.

The Ventilator Breath

5. Pham T, Brochard LJ, Slutsky AS. Mechanical ventilation: state of the art. Mayo Clin Proc 2017; 92:1382–1400.
6. Yang SC, Yang SP. Effects of inspiratory flow waveforms on lung mechanics, gas exchange, and respiratory metabolism in COPD patients during mechanical ventilation. Chest 2002; 122:2096–2104.
7. Kallet RH, Campbell AR, Alonzo JA, et al. The effects of pressure control versus volume control on patient work of breathing in acute lung injury and acute respiratory distress syndrome. Respir Care 2000; 45:1085–1096.

8. Branson RD, Chatburn RL. Controversies in the critical care setting. Should adaptive pressure control modes be utilized for virtually all patients receiving mechanical ventilation? Respir Care 2007; 62:478–485.
9. Singh S, Chien C, Patel S. Pressure regulated volume control (PRVC): set it and forget it? Respir Med Case Rep 2020; 29:100822.
10. Pepe P, Marini JJ. Occult positive end-expiratory pressure in mechanically ventilated patients with airflow obstruction. The auto-PEEP effect. Am Rev Respir Dis 1982; 126:166–170.
11. Hess DR, Kacmarek RM. Basic pulmonary mechanics during mechanical ventilation. In: Essentials of Mechanical Ventilation. New York, McGraw-Hill, 1996:171–176.
12. Tobin MJ. Respiratory monitoring. JAMA 1990; 264:244–251.
13. Bekos V, Marini JJ. Monitoring the mechanically ventilated patient. Crit Care Clin 2007; 23:575–611.
14. Marini JJ. Lung mechanics determinations at the bedside: instrumentation and clinical application. Respir Care 1990; 35:669–696.
15. Katz JA, Zinn SE, Ozanne GM, Fairley BB. Pulmonary, chest wall, and lung–thorax elastances in acute respiratory failure. Chest 1981; 80:304–311.

Assist-Control Ventilation

16. Leung P, Jubran A, Tobin MJ. Comparison of assisted ventilator modes on triggering, patient effort, and dyspnea. Am J Respir Crit Care Med 1997; 155:1940–1948.
17. Laureen H, Pearl R. Flow triggering, pressure triggering, and autotriggering during mechanical ventilation. Crit Care Med 2000; 28:579–581.

Intermittent Mandatory Ventilation

18. Shelledy DC, Rau JL, Thomas Goodfellow L. A comparison of the effects of assist control, SIMV, and SIMV with pressure support on ventilation, oxygen consumption, and ventilatory equivalent. Heart Lung 1995; 24:67–75.
19. Pinsky MR. Cardiopulmonary interactions: physiologic basis and clinical applications. Ann Thorac Soc 2018; 15:S45–S48.
20. Mathru M, et al. Hemodynamic responses to changes in ventilatory patterns in patients with normal and poor left ventricular reserve. Crit Care Med 1982; 10:423–426.

Positive End-Expiratory Pressure

21. Muscedere JG, Mullen JBM, Gan K, Slutsky AS. Tidal ventilation at low airway pressures can augment lung injury. Am J Respir Crit Care Med 1994; 149:1327–1334.
22. Gattinoni L, Protti A, Caironi P, Carlesso E. Ventilator-induced lung injury: the anatomical and physiological framework. Crit Care Med 2010; 38(Suppl):S539–S548.
23. Dreyfuss D, Soler P, Basset G, et al. High inflation pressure pulmonary edema: Respective effects of high airway pressure, high tidal volume, and positive end-expiratory pressure. Am Rev Respir Dis 1988; 137:1159–1164.
24. McKown AC, Semler MW, Rice TW. Best PEEP trials are dependent on tidal volume. Crit Care 2018; 22:115.
25. Takata M, Robotham JL. Ventricular external constraint by the lung and pericardium during positive end-expiratory pressure. Am Rev Respir Dis 1991; 43:872–875.
26. Gainnier M, Michelet P, Thirion X, et al. Prone position and positive end-expiratory pressure in acute respiratory distress syndrome. Crit Care Med 2003; 31:2719–2726.

Chapter 28

The Ventilator-Dependent Patient

Eyes and ears are bad witnesses to men if they have souls that understand not their language.

Heraclitus
6th Century BC

This chapter describes the common practices and daily concerns involved in the care of ventilator-dependent patients. The focus is on artificial airways (endotracheal and tracheostomy tubes), the management of respiratory secretions, and mechanical complications of positive-pressure ventilation (i.e., barotrauma and occult PEEP). The infectious complications of mechanical ventilation are described in the next chapter.

ARTIFICIAL AIRWAYS

Positive pressure ventilation is delivered through a variety of plastic tubes that are passed into the trachea through the vocal cords (endotracheal tubes) or are inserted directly into the trachea (tracheostomy tubes). These tubes are equipped with an inflatable balloon at the distal end (called a cuff) that is used to seal the trachea and prevent positive-pressure inflation volumes from escaping out through the larynx.

Endotracheal Tubes

Endotracheal (ET) tubes vary in length from 25 to 35 cm, and are sized according to their internal diameter, which varies from 5 to 10 mm (e.g., a size 7 ET tube will have an internal diameter of 7 mm). The size of the airways is primarily related to body height, and the recommended size of ET tubes according to height is as follows: size 7 ET tube for height ≤5 feet, 2 inches (5'2"), size 7.5 for height 5'3" to 5'10", and size 8 for height of 5'11'' or taller (1). Despite these recommendations, there is no evidence that ET tube size affects clinical outcomes (2).

Proper Tube Position

Evaluation of ET tube position is mandatory after intubation, and the portable chest film in

Figure 28.1 shows the proper position of an ET tube. When the head is in a neutral position, *the tip of the ET tube should be 3 to 5 cm above the tracheal bifurcation or midway between the bifurcation and vocal cords*. (This bifurcation is often identified as the *carina*, which is a ridge of cartilage situated at the tracheal bifurcation.) If the bifurcation or carina is not visible, it can be located at the level of the T4–T5 interspace on a portable chest x-ray. Flexion or extension of the head and neck causes a 2 cm displacement of the tip of the endotracheal tube (3).

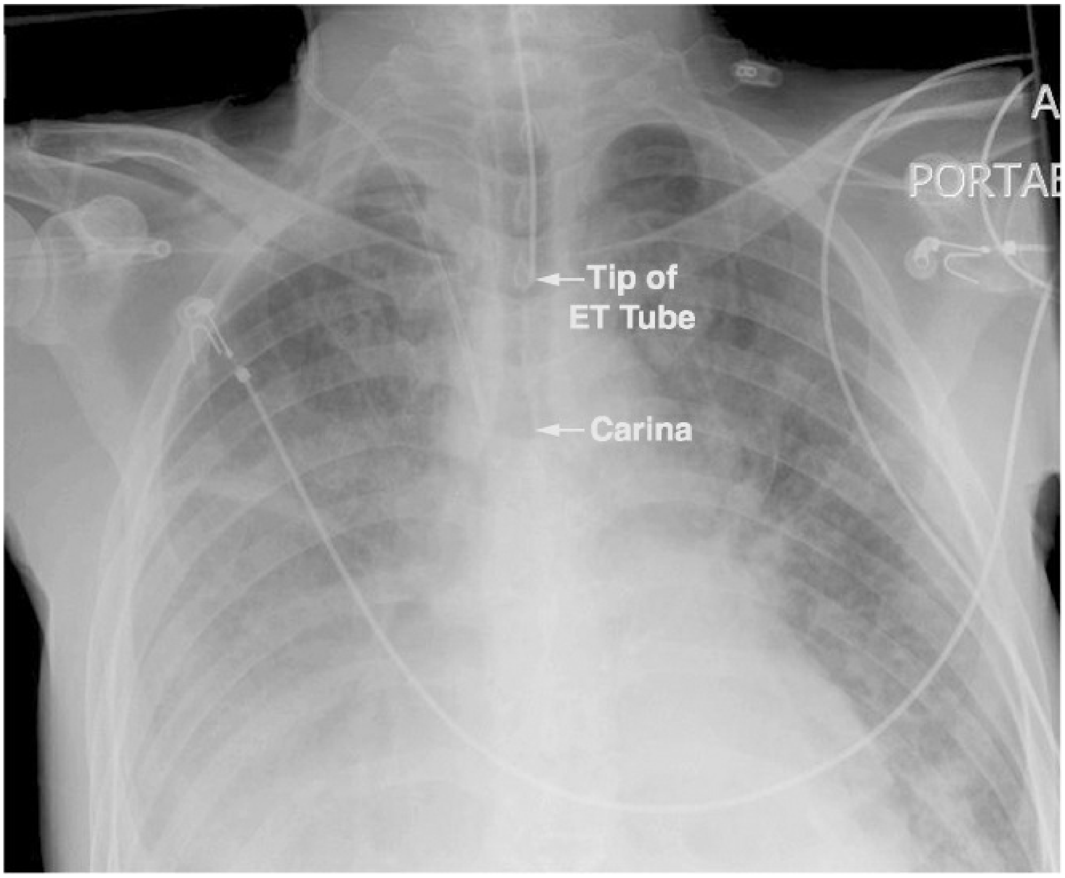

FIGURE 28.1 Portable chest x-ray showing proper position of an endotracheal tube, with the tip of the tube located midway between the thoracic inlet and the tracheal bifurcation (carina).

ESTIMATING LENGTH OF INSERTION: For endotracheal intubation, the proper length of ET tube insertion can be estimated by locating the manubriosternal joint, which forms a convex angle on the chest wall (known as the angle of Louis) that is in the same horizontal plane as the tracheal bifurcation. With the head fully extended, the distance between the upper incisors and the manubriosternal joint is the insertion distance that should place the tip of the ET tube in the appropriate position (4).

Migration of Tubes

ET tubes can migrate in either direction along the trachea (from manipulation of the tube during suctioning, turning, etc.), and when tubes move distally into the lungs, they often end up in the right mainstem bronchus (which runs a straight course down from the trachea). This is demonstrated in Figure 28.2 (5), which shows the tip of the ET tube in the right mainstem bronchus, resulting in atelectasis of the non-ventilated left lung. The risk of ET tube migration can be reduced by keeping the tip of ET tube no further than 21 cm from the teeth in women and 23 cm from the teeth in men (6).

Laryngeal Damage

Laryngeal injury has been demonstrated in more than one-half of ICU patients that are intubated for longer than 12 hours (7), and is more common with difficult intubations, large ET tubes, and prolonged intubations. The spectrum of laryngeal damage includes posterior and subglottic ulceration, granulomas, and laryngeal edema. Long-term problems with voice and shortness of breath have been reported (7), and laryngeal edema can be problematic after extubation (described in Chapter 30).

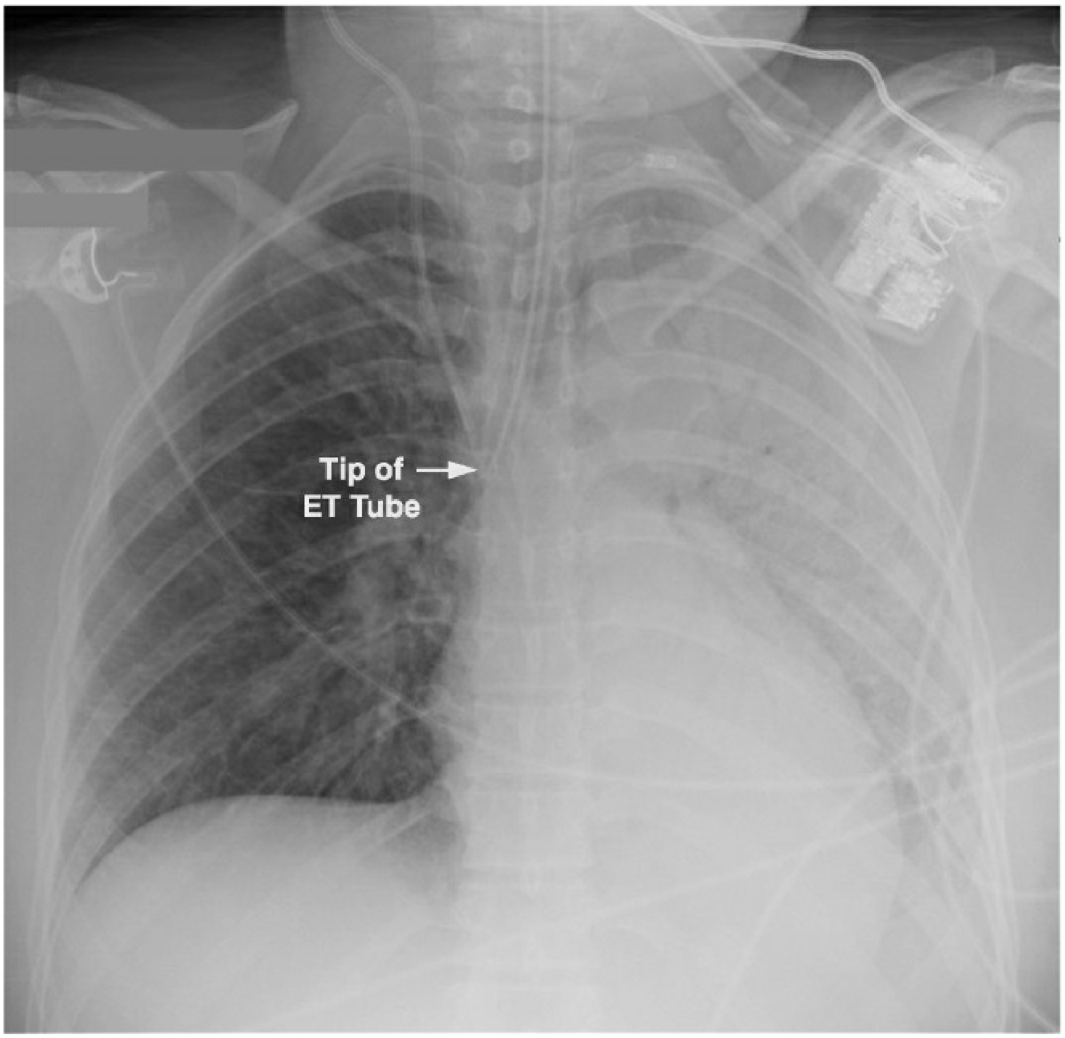

FIGURE 28.2 Portable chest x-ray showing intubation of the right mainstem bronchus and atelectasis of the non-ventilated left lung. Image from Reference 5.

Subglottic Drainage Tubes

The prominent role played by aspiration of mouth secretions in the pathogenesis of ventilator-associated pneumonia has led to the introduction of specially-designed endotracheal tubes capable of draining mouth secretions that accumulate just above the inflated cuff. These tubes can reduce the incidence of ventilator-associated pneumonia (8), and they are described in more detail in the next chapter (see Figure 29.2).

Tracheostomy

Tracheostomy is preferred in patients who require prolonged mechanical ventilation (>2 weeks). The are several advantages with tracheostomy, including greater patient comfort, easier access to the airways for clearing secretions and bronchodilator administration, reduced resistance for breathing, and reduced risk of laryngeal injury.

Tracheostomy Timing

Several studies have compared early tracheostomy (within one week after intubation) with late tracheostomy (two weeks or later after intubation), and the bulk of the evidence shows that early tracheostomy can reduce sedative requirements and reduce the time on mechanical ventilation, but it does not reduce the incidence of ventilator-associated pneumonia, and does not reduce the mortality rate (9–11). The common practice is to *consider tracheostomy after one week of intubation if there is little chance of extubation in the following week.*

Techniques

The traditional method of performing a tracheotomy as an open surgical procedure has been replaced in popularity by *percutaneous dilatational tracheostomy*, where a guidewire is passed through a small needle puncture in the anterior wall of the trachea, and a tracheostomy tube is advanced over the wire and into the tracheal lumen. This technique, which is performed at the bedside, has several advantages over surgical tracheostomy (including less cost, avoiding general anesthesia, and fewer infections) (12), and is the preferred method of tracheostomy (13). However, the percutaneous technique has more technical difficulties than the surgical procedure (12), emphasizing the importance of a skilled and experienced operator.

The technique known as *cricothyroidotomy* is used only for emergency access to the airway. The trachea is entered through the cricothyroid membrane, just below the larynx, and there is high incidence of laryngeal injury and subglottic stenosis. Patients who survive following a cricothyroidotomy should have a regular tracheostomy (surgical or percutaneous) as soon as they are stable.

Complications

Combining surgical and percutaneous tracheostomy, the mortality rate is less than 1%, the incidence of major bleeding is <5%, and the infection rate is 2–10% (with the lower rates in the percutaneous tracheostomies) (12).

ACCIDENTAL DECANNULATION: One acute complication that deserves mention is accidental decannulation. If the tracheostomy tube is dislodged before the stoma tract is mature (which takes about one week) the tract closes quickly, and blind reinsertion of the tube can create false tracts. *If a tracheostomy tube is dislodged within a few days of insertion, the patient should be reintubated orally before attempting to reinsert the tracheostomy tube*.

TRACHEAL STENOSIS: The most feared complication of tracheostomy is tracheal stenosis, which is a late complication that appears in the first 6 months after the tracheostomy tube is removed. Most cases of tracheal stenosis occur at the site of the tracheal incision, and are the result of tracheal narrowing after the stoma closes. The incidence of tracheal stenosis ranges from zero to 15% in individual reports (14), but most cases are asymptomatic. The risk of tracheal stenosis is the same with surgical and percutaneous tracheostomies (15).

Cuff Management

Positive pressure ventilation requires a seal in the trachea that prevents gas from escaping out through the larynx during lung inflation, and this seal is created by inflatable balloons (called cuffs) that surround the distal portions of endotracheal and tracheostomy tubes. A tracheostomy tube with an inflated cuff is shown in Figure 28.3. The cuff is attached to a pilot balloon that has a one-way valve. A syringe is attached to the pilot balloon and air is injected into the cuff until there is no audible leak around the cuff. (The pilot balloon will inflate as the cuff inflates.)

The pressure in the cuff (measured with a pressure gauge attached to the pilot balloon) should not exceed 25 mm Hg (16). This pressure limit is based on the assumption that the capillary hydrostatic pressure in the wall of the trachea is 25 mm Hg, and thus external (cuff) pressures above 25 mm Hg can compress the underlying capillaries and produce ischemic injury in the tracheal mucosa. Modern cuffs have an elongated design that reduces the pressure needed for a tracheal seal.

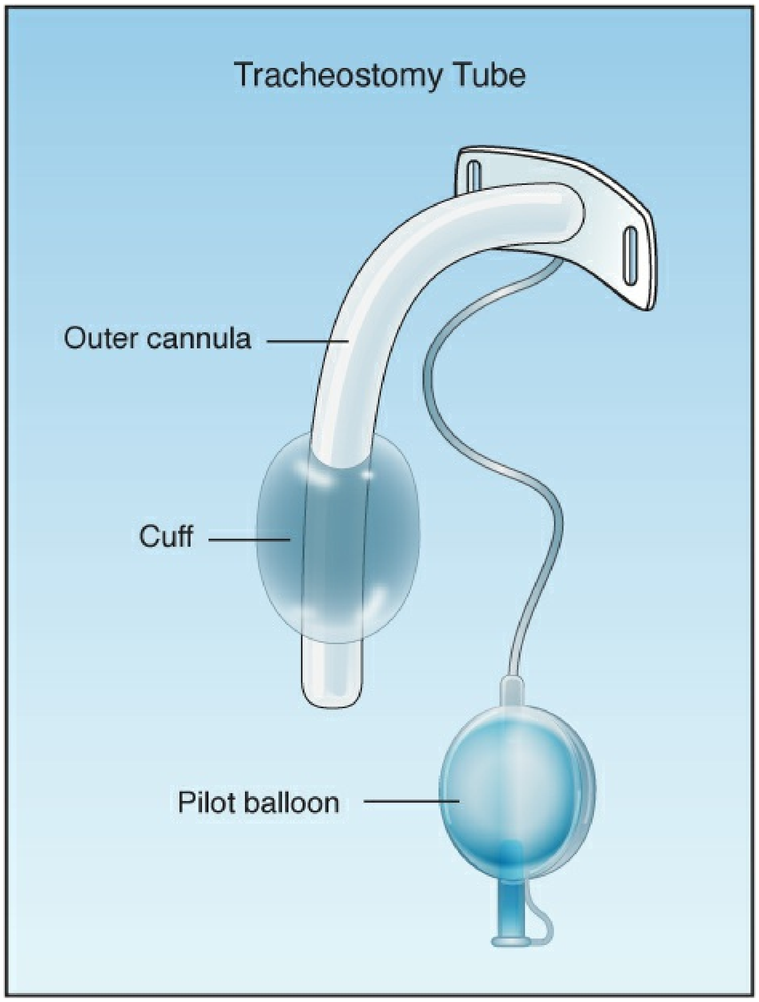

FIGURE 28.3 A tracheostomy tube with an inflated cuff.

Cuff Leaks

Cuff leaks are usually detected by audible sounds during lung inflation (created by gas flowing through the vocal cords). When a cuff leak becomes audible, the volume of the leak can be estimated as the difference between the desired tidal volume and the exhaled tidal volume measured by the ventilator. Cuff leaks are usually not caused by disruption of the cuff (17), but are instead the result of: (*a*) nonuniform contact between the cuff and the wall of the trachea, or (*b*) dysfunction of the valve on the pilot balloon.

TROUBLESHOOTING: If a cuff leak is apparent, the patient should be separated from the ventilator and the lungs should be manually inflated with an anesthesia bag. Then check the pilot balloon:

. If the pilot balloon is inflated, then the cuff is inflated, and the problem is likely to be a non-uniform seal with the wall of the trachea. In this situation, deflate the cuff, and reposition the tube, then re-inflate the cuff. (You can put more air into the cuff, but make sure the pressure does not exceed 25 cm H_2O). If this does not correct the leak, then replace the tube with a larger one.

. If the pilot ballon is deflated, then the likely problem is a faulty one-way valve on the pilot balloon, or a disrupted cuff. In this situation, attempt to re-inflate the cuff, and if the pilot balloon does not inflate, then replace the tube.

AIRWAY CARE

The clearance of respiratory secretions in intubated patients is often referred to as “pulmonary toilet”. However, because the lungs are not a toilet, the term “airway care” seems more appropriate.

Suctioning

Aspiration of airway secretions is a standard practice in the care of ventilator-dependent patients. However, this can have a number of unfavorable effects, including increases in heart rate, blood pressure, intracranial pressure, and occult PEEP (described later), and a decrease in arterial O_2 saturation (18). More importantly, biofilms containing pathogenic organisms are found on the inner surface of ET tubes and tracheostomy tubes (see Figure 28.4), and passing a suction catheter through the tubes can dislodge these biofilms and inoculate the lungs with pathogenic organisms (19). As a result of these risks, *clinical practice guidelines do not recommended routine suctioning of the airways* (18). Instead, suctioning should be performed only when there is evidence of respiratory secretions (see next).

Indications

The following are considered the best indicators for airway suctioning (18,20): (*a*) visible secretions in airway tubes or ventilator tubing, (*b*) auscultatory evidence of airway secretions, and (*c*) a sawtooth pattern on the ventilator flow waveform. Other indications could include x-ray evidence of segmental atelectasis and an abrupt decrease in arterial oxygenation, which could be the result of retained secretions.

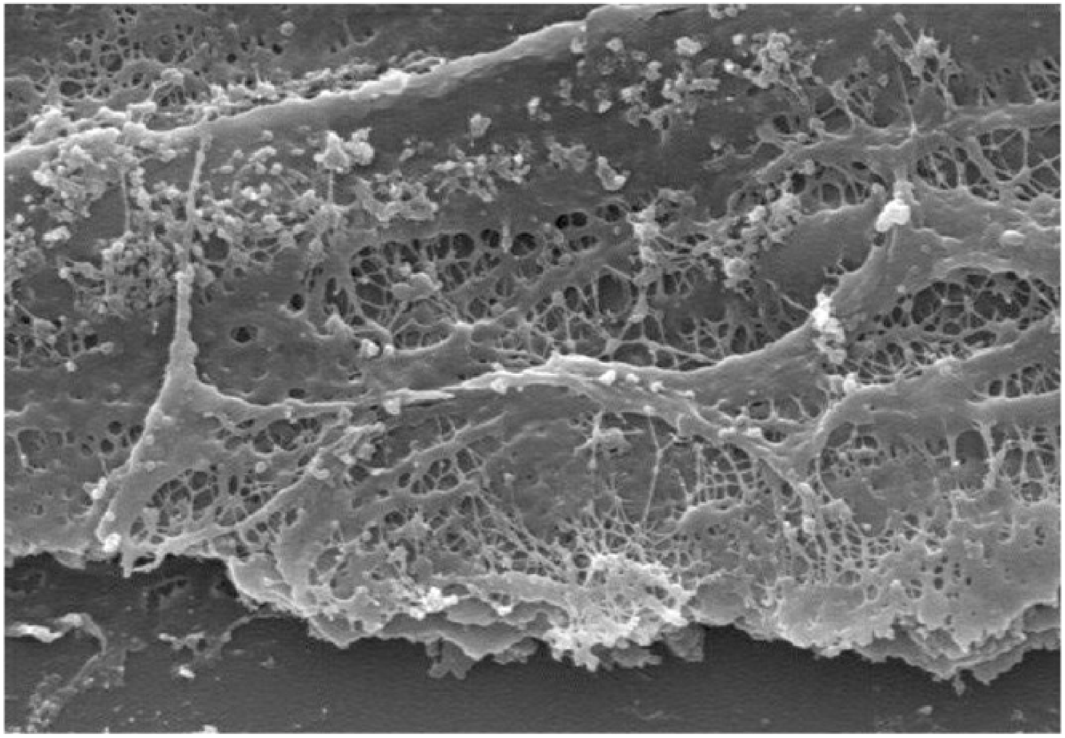

FIGURE 28.4 Electron micrograph showing a biofilm on the inner surface of an endotracheal tube. Image from Reference 19.

Saline Instillation

Once a standard practice, *the routine use of saline to facilitate the aspiration of secretions is NOT recommended* (18). There are two reasons for this: (*a*) saline will not reduce the viscosity of respiratory secretions (as described next), and (*b*) injection of saline can dislodge pathogenic organisms colonizing the inner surface of airway tubes. One study has shown that *injection of 5 mL of saline can dislodge up to 300,000 colonies of viable bacteria from the inner surface of endotracheal tubes* (21).

Viscosity of Respiratory Secretions

The respiratory secretions create a blanket that covers the mucosal surface of the airways. This blanket has a hydrophilic (water soluble) layer that faces inward to keep the mucosal surface moist, and a hydrophobic (water insoluble) layer that faces outward to trap particles and debris in the airways. This trapping is achieved by a meshwork of mucoprotein strands that are held together by disulfide bridges, and the combination of the mucoprotein meshwork and the trapped debris determines the viscoelastic behavior of the secretions. Thus, the layer that contributes to the viscosity of respiratory secretions is not water soluble, and hence *saline will not reduce the viscosity of respiratory secretions*. Adding saline to viscous respiratory secretions is like pouring water over grease.

Mucolytic Therapy

When respiratory secretions are thick and tenacious, a mucolytic agent like *N-Acetylcysteine* can help to clear the airways. N-acetylcysteine (NAC) is a sulfhydryl-containing tripeptide that is better known as the antidote for acetaminophen overdose, but it is also a mucolytic agent that acts by disrupting the disulfide bridges between mucoprotein strands in respiratory secretions (22). The drug is available in a liquid preparation (10 or 20% solution) that can be given as an aerosol spray, or injected directly into the airways (Table 28.1). Aerosolized NAC is irritating and can provoke coughing and bronchospasm (especially in asthmatics), and direct instillation of NAC into the airway should be preferred for clearing secretions in ventilator-dependent patients.

TABLE 28.1 Mucolytic Therapy with N-Acetylcysteine (NAC)

Method	Regimen
Aerosol Therapy	• Use 10% NAC solution. • Mix 2.5 mL NAC with 2.5 mL saline and place mixture (5 mL) in a small volume nebulizer for aerosol delivery. Can repeat every 8 hrs. • Warning: Aerosol NAC can provoke bronchospasm, and is not recommended in asthmatics.
Tracheal Injection	• Use 20% NAC solution. • Mix 1 mL NAC with 1 mL saline and inject (2 mL) into the trachea. Can repeat every 8 hrs. • Warning: Prolonged use (>48 hrs) can promote bronchorrhea.

PULMONARY BAROTRAUMA

As described in Chapter 24, one of the principal mechanisms of ventilator-induced lung injury is overdistension of alveoli in normal lung regions (23), which ruptures the alveolar-capillary interface (see Figure 24.6). This is known as *volutrauma*, and it produces an inflammatory infiltration in the lung that is very similar to the acute respiratory distress syndrome (ARDS). Another consequence of alveolar rupture is the escape of gas from the distal airspaces, but this is called *pulmonary barotrauma*. If you are confused about the terms volutrauma and barotrauma, consider that *volutrauma is an expression of the strain (i.e., degree of deformation) on alveoli, while barotrauma is an expression of stress (i.e., the forces that produce deformation) imposed on alveoli, which is the transpulmonary pressure* (24).

Clinical Presentation

Escape of gas from the alveoli can produce a variety of clinical manifestations. The alveolar gas can dissect along tissue planes and produce *pulmonary interstitial emphysema*, and can move into the mediastinum and produce *pneumomediastinum*. Mediastinal gas can move into the neck to produce *subcutaneous emphysema*, or can pass below the diaphragm to produce *pneumoperitoneum*. Finally, if the rupture involves the visceral pleura, gas will collect in the pleural space and produce a *pneumothorax*. Each of these entities can occur alone or in combination with the others.

Pneumothorax

The incidence of pneumothorax is 3–10% in ventilator-dependent patients who receive limited-volume "lung protective ventilation" (25), and it has been reported in 35% of patients with COVID-19 pneumonia (mode of ventilation unknown) (26). In addition to barotrauma, iatrogenic pneumothoraces can be the result of procedures (i.e., insertion of central venous catheters, thoracentesis, and bronchoscopy).

Clinical Presentation

Pneumothoraces from procedures can be small and clinically silent, while those caused by barotrauma are usually larger and can cause hypoxemia and even hypotension (see later). *The most specific clinical finding is subcutaneous emphysema*, which appears initially in the neck and upper thorax. The absence of breath sounds is an unreliable finding in ventilator-dependent patients because sounds transmitted from the tracheal tubes and ventilator tubing can be mistaken for airway sounds.

Radiographic Detection

The radiographic *detection of pleural air can be difficult in the supine position, because pleural air does not collect at the lung apex when patients are supine* (27). This is illustrated in Figure 28.5. In this case of a traumatic pneumothorax, the chest x-ray is unrevealing, but the CT scan reveals an anterior pneumothorax on the left. Pleural air will collect in the most superior region of the hemithorax, which in the supine position, is the region just anterior to both lung bases. Therefore, *basilar and subpulmonic collections of air are characteristic of pneumothorax in the supine position* (27).

Ultrasound Detection

Ultrasound of the lungs has consistently proven to be more sensitive than chest radiography for the detection of pneumothorax (28), but the yield is highly dependent on the skill and experience of the operator.

METHOD: The linear ultrasound probe is placed on the anterior chest and perpendicular to the 3rd or 4th intercostal space along the midclavicular line. The goal is to identify the pleural line (see Figure 28.6), a bright, hyperechoic line that represents the apposition of the parietal and visceral pleura. Normally, the visceral pleura moves along the parietal pleura as the lung inflates and deflates, and this sliding movement creates a shimmering effect on the ultrasound image. When air is present in the pleural space, the two pleural surfaces separate, and the sliding movement of the pleural line is lost. The absence of "lung sliding" is therefore suggestive (but not pathognomonic) of pneumothorax (29). (However, the presence of lung sliding rules out the diagnosis of pneumothorax.)

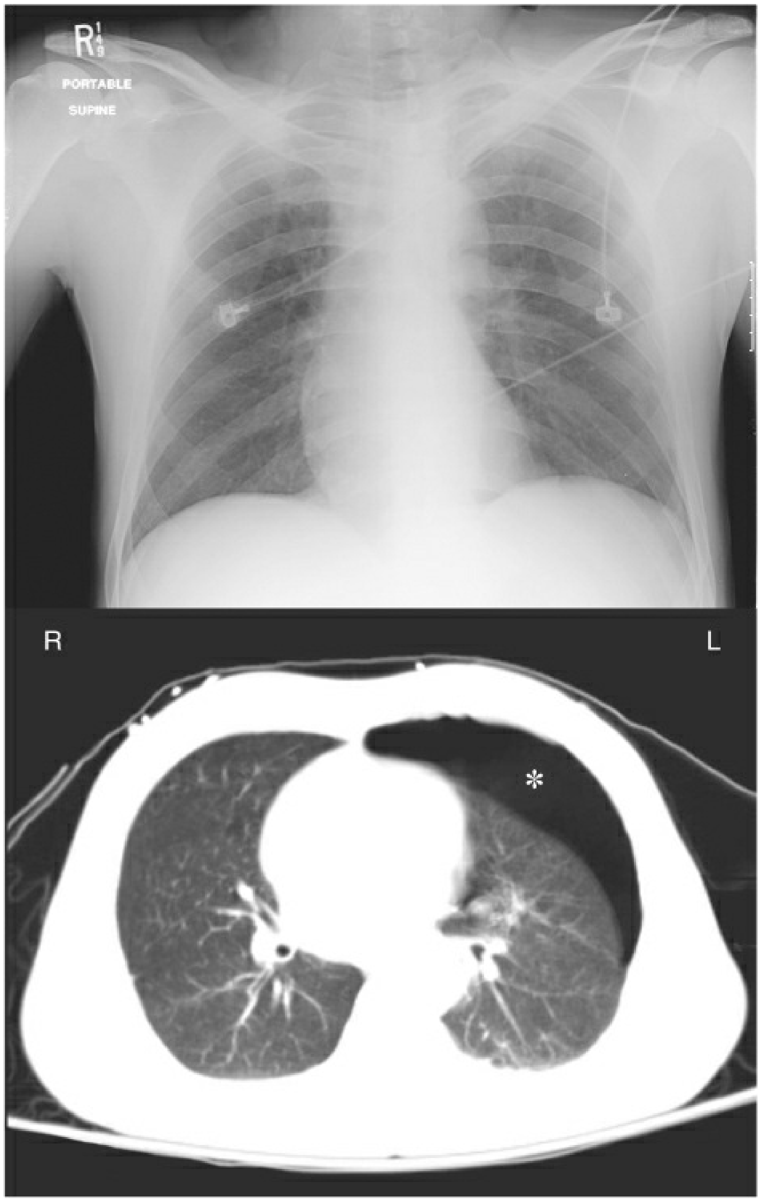

FIGURE 28.5 A portable chest x-ray and CT image of the thorax in a young male with blunt trauma to the chest. An anterior pneumothorax is evident on the CT image (indicated by the asterisk) but is not apparent on the portable chest x-ray.

When lung sliding is absent, the ultrasound probe should be moved laterally and inferiorly in increments, to reach the point where the two pleural surfaces meet again (at the edge of the pneumothorax), which is indicated by the return of lung sliding. This is known as the "lung point", and it is pathognomonic of pneumothorax (29).

Tension Pneumothorax

Pleural air can accumulate rapidly when a pneumothorax is the result of positive pressure ventilation, and this accumulation can eventually produce a life-threatening *tension pneumothorax* where the increase in intrathoracic pressure collapses the ipsilateral lung, and causes a progressive decrease in cardiac output that culminates in "obstructive shock", and cardiac arrest from pulseless electrical activity. This condition should be considered in any ventilator-dependent patient who develops unexplained hypoxemia or hypotension. It is easily recognized on a portable chest x-ray, as shown in Figure 28.7 (30). Note the absence of lung markings in the left hemothorax (indicating complete collapse of the left lung), and the mediastinal shift to the right (indicating a marked increase in pressure in the left hemithorax. The left hemidiaphragm is also depressed, which is another sign of increased intrapleural pressure.

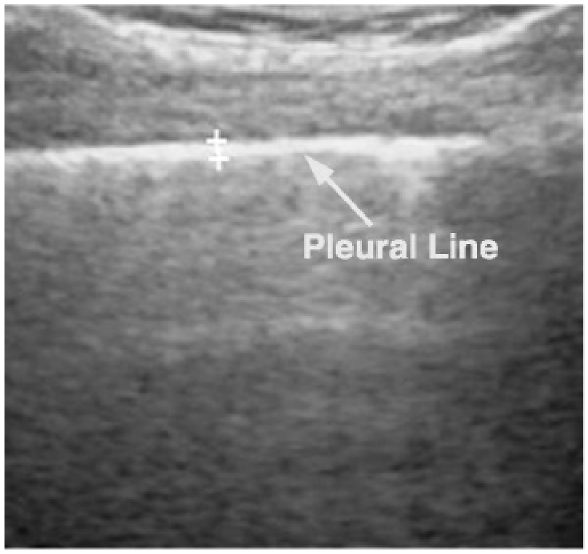

FIGURE 28.6 Ultrasound image showing the hyperechoic pleural line, which represents the apposition of the visceral and parietal pleura. A sliding movement in the pleural line with respirations occurs normally, and the absence of “lung sliding is suggestive of a pneumothorax. See text for further explanation.

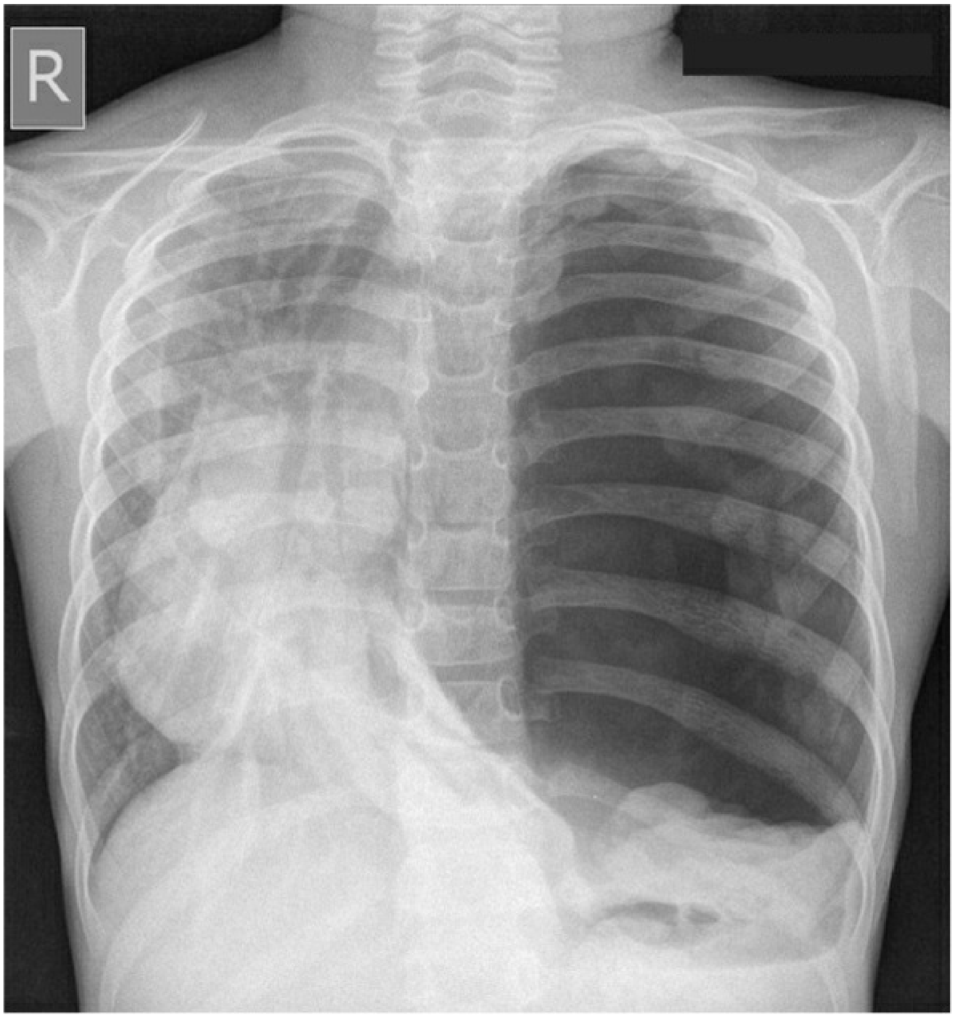

FIGURE 28.7 Tension pneumothorax. Note the absence of lung markings in the left hemithorax, with depression of the left hemidiaphragm and contralateral shift of the mediastinum. X-ray from Reference 30.

DECOMPRESSION: Tension pneumothorax is a medical emergency, and requires immediate decompression. The recommended procedure is insertion of a 14-gauge, 8 cm (3.25 inch) angiocatheter in the fifth intercostal space just anterior to the midaxillary line (31,32). (The failure rate is higher when shorter catheters are used, and when the catheters are inserted in the second interspace at the midclavicular line).

Needle decompression is not always successful, either because the needle does not reach the pleural space, or because the catheters kink or are dislodged. If this is the case, then "finger decompression" is recommended (i.e., placing a finger in the pleural space) (31). Following decompression, a small-bore chest tube (8.5–14 French) should be inserted.

Pleural Evacuation

Evacuation of pleural air is accomplished by inserting a chest tube through the fourth or fifth intercostal space along the mid axillary line. The tube should be advanced in an anterior and superior direction to evacuate air (and is directed inferiorly to evacuate fluid). Pleural evacuation is achieved with a three-chamber system, which is depicted in Figure 28.8.

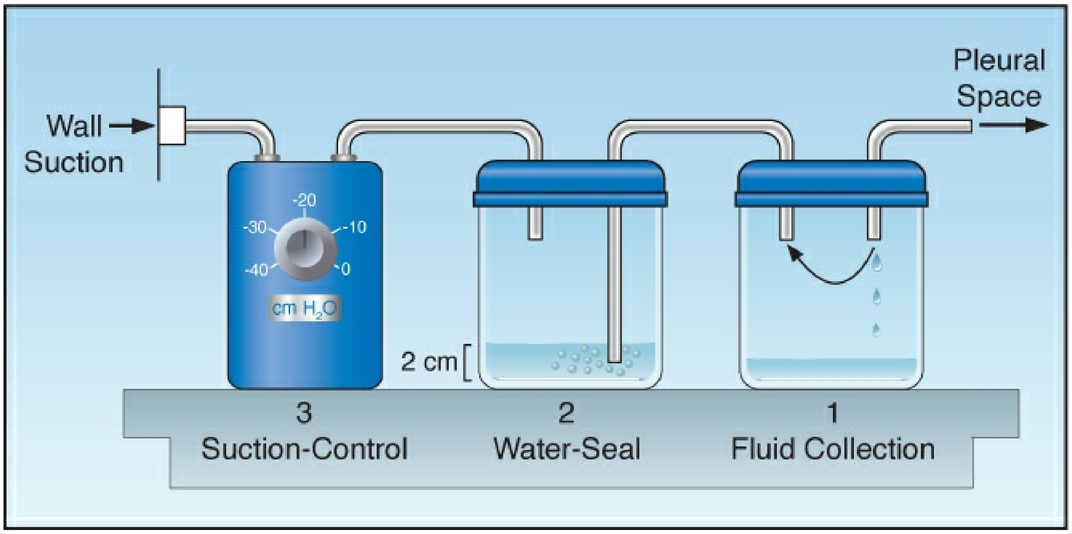

FIGURE 28.8 A three-compartment pleural drainage system for evacuating air and fluid from the pleural space.

Fluid Collection

The first chamber in the system collects fluid from the pleural space and allows air to pass through to the next chamber. Because the inlet of this chamber is not in direct contact with the fluid, the accumulating fluid does not exert a back pressure on the pleural space.

Water Seal Chamber

The second chamber creates a one way valve for the pleural space using a column of water (usually 2 cm in height) to produce a slightly positive pleural pressure (2 cm H_2O) that prevents atmospheric air from entering the pleural space. The water thus "seals" the pleural space from the surrounding atmosphere, hence this is called the "water seal chamber".

BUBBLING: Air that is evacuated from the pleural space will pass through the water and create bubbles. Thus, *the presence of bubbles in the water seal chamber is evidence of an air leak through a bronchopleural fistula (a hole in the lungs).*

Suction Control

The final component of the system is an adjustable valve that is connected to wall suction and controls the level of negative pressure used for pleural drainage. This pressure is usually set at –20 cm H_2O.

Why Suction?

The practice of using suction to evacuate pleural air is often unnecessary, and can be counterproductive. Suction may help to reinflate the lungs when the pleural air is evacuated, but thereafter the negative pressure in the pleural space will increase the transpulmonary pressure (the pressure difference between alveoli and the pleural space), and this will promote the flow of air through a bronchopleural fistula. Thus *applying suction to the pleural space will promote air* *leaks from the lungs and prevent the closure of bronchopleural fistulas*. If there is a persistent air leak when suction is being applied to the pleural space, the suction should be turned off. Air leaks from the lungs will continue to be evacuated when the pleural pressure becomes more positive than the water seal pressure.

OCCULT (INTRINSIC) PEEP

Occult PEEP (also called intrinsic PEEP and auto-PEEP) is the result of incomplete emptying of the lungs. It is common in patients with severe airflow obstruction (e.g., it is responsible for the hyperinflation in acute asthma), and is exacerbated by rapid breathing and high tidal volumes (33). In ventilator-dependent patients, occult PEEP is considered universal in those with asthma and COPD (34,35), and low levels (≤3 cm H_2O) are also common in patients with ARDS (36).

Adverse Effects

Occult PEEP has several adverse effects, which are summarized below (33).

- Occult PEEP increases mean intrathoracic pressure, which can impair venous return and decrease cardiac output.
- The hyperinflation associated with occult PEEP increases the work of breathing because greater negative pressures are required to draw the tidal volume into the lungs. (To appreciate this effect, take a deep breath, and then try breathing from there.)
- Occult PEEP increases end-inspiratory alveolar volume and pressure, which increases the risk of volutrauma and barotrauma.
- Occult PEEP can be transmitted into the superior vena cava and create a spurious increase in central venous pressure measurements.

Monitoring Occult PEEP

As indicated by its name, occult PEEP is not evident in the airway pressures monitored during mechanical ventilation. It is easy to detect but difficult to quantify. Evidence of occult PEEP is provided by the expiratory flow waveform, which shows the presence of airflow at the end of expiration (see Figure 23.5). Determining the level of occult PEEP requires an end-expiratory occlusion maneuver.

End-Expiratory Occlusion

Occlusion of the expiratory circuit at the end of expiration will reveal occult PEEP, because in the absence of airflow, the pressure in the proximal airways will be equivalent to end-expiratory alveolar pressure. This is demonstrated in Figure 28.9. Accuracy requires that the occlusion occur at the very end of expiration, and this cannot be timed properly if patients are breathing spontaneously. Therefore, *the end-expiratory occlusion method is reliable only in patients who are not breathing spontaneously*.

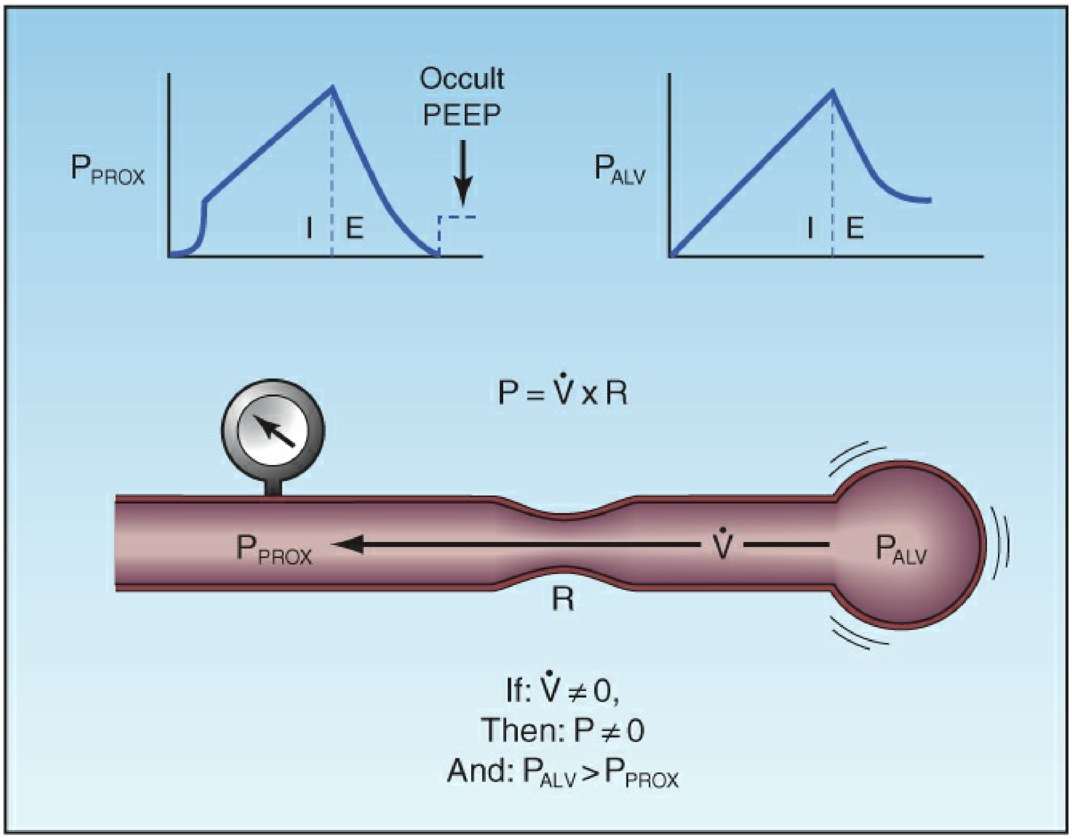

FIGURE 28.9 The features of occult PEEP. The presence of airflow (V) at end expiration indicates a pressure drop from the alveolus (PALV) to the proximal airways (PPROX). As shown at the top of the figure, the proximal airway pressure returns to zero at end expiration while the alveolar pressure remains positive; hence the term *occult* PEEP. The upper left panel illustrates the end expiratory occlusion method for measuring the level of occult PEEP.

Management

The maneuvers used to prevent or reduce hyperinflation and occult PEEP are all directed at promoting alveolar emptying during expiration. These maneuvers include reducing the tidal volume, increasing the inspiratory flow rate, reducing the inspiratory time (in pressure control ventilation), and reducing the respiratory rate, if possible. Some of these maneuvers will reduce the minute ventilation, which may not be desirable.

Applied PEEP

The addition of external PEEP can reduce hyperinflation (and occult PEEP) by holding the small airways open at end-expiration. The level of applied PEEP must be enough to counterbalance the pressure causing small airways collapse, but should not exceed the level of occult PEEP (so that it does not impair expiratory flow) (37). To accomplish this, the level of applied PEEP should be just below the level of occult PEEP. Since occult PEEP is difficult to quantify in spontaneously breathing patients, an alternative method is to monitor the level of end-expiratory airflow in response to applied PEEP; i.e., if the applied PEEP reduces or eliminates end-expiratory flow, then it is reducing the level of occult PEEP. However, the end result is still PEEP (applied PEEP instead of occult PEEP), but the applied PEEP will help to reduce the risk of atelectrauma from repetitive opening and closing of small airways at the end of expiration. (*Note:* The effect of applied PEEP on occult PEEP is included for completeness, but attempting to reduce occult PEEP with applied PEEP is probably not worth the trouble.)

A FINAL WORD

The patient's clinical course in the first few days of mechanical ventilation will give you an indication of what is coming. If the patient is not improving after the first week, proceed to tracheostomy (for patient comfort and improved clearance of secretions). Most of the day-to-day management of the ventilator-dependent patient involves vigilance for adverse events (i.e., "putting out fires"). You will learn that, in many cases, you are not the one controlling the course of the patient's illness.

References

Endotracheal Tubes

1. Cao AC, Rereddy S, Mirza N. Current practices in endotracheal tube size selection in adults. Laryngoscope 2021; 131:1967–1971.
2. Esianor BI, Campbell BR, Casey JD, et al. Endotracheal tube size in critically ill patients. JAMA Otolaryngol Head Neck Surg 2022; 148:849–853.
3. Goodman LR. Pulmonary support and monitoring apparatus. In: Goodman LR, Putman CE, eds. *Critical Care Imaging*. 3rd ed. Philadelphia: WB Saunders, 1992; 35–59.
4. Lee B-J, Yi J-W, Chung JY, et al. Bedside prediction of airways length in adults and children. Anesthesiology 2009; 111:556–560.
5. Feng K. Right main bronchus intubation causing left lung atelectasis. Case study. Radiopaedia.org. (Accessed Feb 11, 2024)
6. Owen RL, Cheney FW. Endotracheal intubation: a preventable complication. Anesthesiology 1987; 67:255–257.
7. Shinn JR, Kimura KS, Campbell BR, et al. Incidence and outcomes of acute laryngeal injury after prolonged mechanical ventilation. Crit Care Med 2019; 47:1699–1706.
8. Muscedere J, Rewa O, McKechnie K, et al. Subglottic secretion drainage for the prevention of ventilator-associated pneumonia: a systematic review and meta-analysis. Crit Care Med 2011; 39:1985–1991.

Tracheostomy

9. Deng H, Fang Q, Chen K, Zhang X. Early versus late tracheostomy in ICU patients: a meta-analysis of randomized controlled trials. Medicine 2021; 100:e24329.
10. Meng L, Wang C, Li J, et al. Early versus late tracheostomy in critically ill patients; a systematic review and meta-analysis. Clin Resp J 2016; 10:684–692.
11. Terragni PP, Antonelli M, Fumagalli R, et al. Early vs late tracheotomy for prevention of pneumonia in mechanically ventilated adult ICU patients. JAMA 2010; 303:1483–1489.
12. Putensen C, Theuerkauf N, Guenther U, et al. Percutaneous and surgical tracheostomy in critically ill adult patients: A meta analysis. Crit Care 2014; 18:544.
13. Raimondi N, Vidal MR, Calleja J, et al. Evidence based guidelines for the use of tracheostomy in critically ill patients. J Crit Care 2017; 38:304–318.
14. Heffner JE. Tracheotomy: application and timing. Clin Chest Med 2003; 24:389–398.
15. Freeman BD, Morris PE. Tracheostomy practice in adults with acute respiratory failure. Crit Care Med 2012; 40:2890–2896.
16. Heffner JE, Hess D. Tracheostomy management in the chronically ventilated patient. Clin Chest Med 2001; 22:5; 10:561–568.
17. Kearl RA, Hooper RG. Massive airway leaks: an analysis of the role of endotracheal tubes. Crit Care Med 1993; 21:518–521.

Airway Care

18. Blakeman TC, Scott JB, Yoder MA, et al. AARC clinical practice guidelines: artificial airway suctioning. Respir Care 2022; 67:258–271.
19. Gil-Perontin S, Ramirez P, Marti V, et al. Implications of endotracheal tube biofilm in ventilator associated pneumonia response: a state of concept. Crit Care 2012; 16:R93.
20. Guglielminotti J, Alzieu M, Maury E, et al. Bedside detection of retained tracheobronchial secretions in patients receiving mechanical ventilation: is it time for tracheal suctioning? Chest 2000; 118: 1095–1099.

21. Hagler DA, Traver GA. Endotracheal saline and suction catheters: sources of lower airways contamination. Am J Crit Care 1994; 3:444–447.
22. Sathe NA, Krishnaswami S, Andrews J, et al. Pharmacological agents that promote airway clearance in hospitalized subjects: a systematic review. Respir Care 2015; 60:1061–1070.

Pulmonary Barotrauma

23. Slutsky AS, Ranieri VM. Ventilator-induced lung injury. N Engl J Med 2013; 369:2126–2136.
24. Gattinoni L, Marini JJ, Collino F, et al. The future of mechanical ventilation: lessons from the present and the past. Crit Care 2017; 21:183.
25. Anzueto A, Frutos-Vivar F, Esteban A, et al. Incidence, risk factors and outcomes of barotrauma in mechanically ventilated patients. Intensive Care Med 2004; 30:612–619.
26. Khan S, Hilal K, Arshad A, et al. Pulmonary barotrauma as a complication of mechanical ventilation for management of COVID-19 associated acute respiratory distress syndrome (CARDS). J Pak Med Assoc 2023; 74:43–47.
27. Tocino IM, Miller MH, Fairfax WR. Distribution of pneumothorax in the supine and semirecumbent critically ill adult. Am J Radiol 1985; 144:901–905.
28. Ron E, Alattar Z, Hoebee S, et al. Current trends in the use of ultrasound to identify pneumothoraces in ICU, Trauma, and ARDS patients. J Intensive Care Med 2022; 37:5–11.
29. Volpicelli G. Sonographic diagnosis of pneumothorax. Intensive Care Med 2011; 37:224–232.
30. Foladi N. Tension pneumothorax. Case study. Radiopaedia.org. (Accessed February 13, 2024.)
31. *ATLS: Advanced Trauma Life Support Student Course Manual*. 10th ed. Chicago: American College of Surgeons, 2018;65–66.
32. Leatherman ML, Held JM, Fluke LM, et al. Relative device stability of anterior versus axillary needle decompression for tension pneumothorax during casualty movement: preliminary analysis of a human cadaver model. J Trauma Acute Care Surg 2017; 83(Suppl 1):S136–S141.

Occult PEEP

33. Marini JJ. Dynamic hyperinflation and auto-positive end-expiratory pressure. Am J Respir Crit Care Med 2011; 184:756–762.
34. Blanch L, Bernabe F, Lucangelo U. Measurement of air trapping, intrinsic positive end-expiratory pressure, and dynamic hyperinflation in mechanically ventilated patients. Respir Care 2005; 50:110–123.
35. Shapiro JM. Management of respiratory failure in status asthmaticus. Am J Respir Med 2002; 1:409–416.
36. Hough CL, Kallet RH, Ranieri M, et al. Intrinsic positive end-expiratory pressure in Acute Respiratory Distress Syndrome (ARDS) Network subjects. Crit Care Med 2005; 33:527–532.
37. Tobin MJ, Lodato RF. PEEP, auto PEEP, and waterfalls. Chest 1989; 96:449–451.

Chapter 29

Ventilator-Associated Pneumonia

Everything hinges on the matter of evidence.

Carl Sagan

Pneumonias that develop during mechanical ventilation can be characterized by one word: *problematic*. The problems include the nonspecific clinical presentation of ventilator-associated pneumonia, the uncertainties in identifying the culprit organism, and the difficulties created by the emergence of multidrug resistant organisms in this condition.

This chapter addresses the multiple problems encountered in the approach to ventilator-associated pneumonia, and includes recommendations from the most recent clinical practice guidelines on the subject (1,2).

GENERAL INFORMATION

The following statements include some basic information about ventilator-associated pneumonia (VAP).

. ICU-acquired pneumonias are divided into ventilator-associated pneumonia (VAP), with onset ≥48 hours after intubation, and hospital-acquired pneumonia (HAP), with onset ≥48 hours after ICU admission in nonintubated patients. VAP is further classified as early onset (2–4 days after intubation) or late-onset (≥5 days after intubation).

. VAP accounts for 80% of ICI-acquired pneumonias, and 60% of VAPs are late-onset.

. The predominant pathogens in VAP and HAP are gram-negative enteric organisms and staphylococci (see Table 29.1) (3). About one-quarter of the infections are polymicrobial.

. Multidrug-resistant pathogens are a growing concern, and are more likely to appear in patients who have received intravenous antibiotics in the past 90 days, in patients who are immunocompromised or receiving renal replacement therapy, and in patients with septic shock (2).

. Viruses are implicated in about 20% of cases of VAP and HAP (4). *Candida* species are common isolates in respiratory secretions, but are not known to cause pneumonia (5).

. VAP prolongs the duration of mechanical ventilation, and increases the length of stay in the ICU (5), but the impact of VAP on survival is unclear (6). Some studies show that VAP has little of no effect on mortality rate (7,8), while others show a small but significant increase in mortality rate (3).

PREVENTIVE MEASURES

Aspiration of microbes from the oropharynx is the inciting event in most cases of VAP (9). Gram-negative aerobic bacilli are the predominant pathogens in the oropharynx of ICU patients (see Figure 4.4), and they are also the predominant pathogens in VAP, as shown in Table 29.1 (3). Early cases of VAP (i.e., appear 2–4 days after intubation) are most likely caused by pathogens that are dragged into the airways during intubation.

TABLE 29.1 Common Isolates in ICU-Acquired Pneumonia

Pathogens	Early VAP[a]	Late VAP[b]	HAP[c]
Enterobacteriaceae	32%	34%	31%
P. aeruginosa	14%	20%	16%
Acinetobacter spp.	8%	21%	13%
Staphylococcus aureus	30%	21%	20%
MSSA	21%	9%	6%
MRSA	9%	13%	13%
Polymicrobial	26%	24%	21%
None identified	25%	22%	38%

From Reference 3. MSSA = methicillin-sensitive *S. aureus*, MRSA = methicillin-resistant *S. aureus.*

[a] Onset 2–4 days after intubation.

[b] Onset ≥5 days after intubation.

[c] Onset ≥48 hrs after ICU admission in non-ventilated patients.

Oral Decontamination

The discovery that pathogens in the oropharynx are the culprits in VAP led to the introduction of oral decontamination as a preventive measure for VAP. Oral decontamination is described in Chapter 4, and is briefly reviewed here.

Methods

There are two methods for eliminating pathogens from the oropharynx (see Table 4.1). The popular (but problematic) method is topical application of 0.12% chlorhexidine gluconate to the oral mucosa every 6 hours. The problem is evidence of increased mortality associated with chlorhexidine (10), which is attributed to a chlorhexidine-induced mucositis that disrupts the mucosal barrier in the mouth and leads to septicemia with pathogenic mouth organisms (11). Because of this risk, *chlorhexidine is no longer recommended for oral decontamination* in the most recent guidelines on VAP (3).

The preferred decontamination method is *selective oral decontamination* (SOD) (3), which uses nonabsorbable antibiotics applied to the oral mucosa (see Table 4.1 for an example of an SOD regimen). The aim of SOD is to “selectively” eliminate pathogens while retaining the normal microbial environment in the mouth (which presumably helps to prevent further colonization with pathogens). The success of this approach is demonstrated in Figure 29.1, which shows that SOD is associated with a significant decrease in both tracheal colonization and pneumonia in ventilator-dependent patients (12). Unfortunately, SOD has been unpopular in the United States, primarily due to the unfounded fear that SOD will promote bacterial resistance. (See Chapter 4 for more on this topic.)

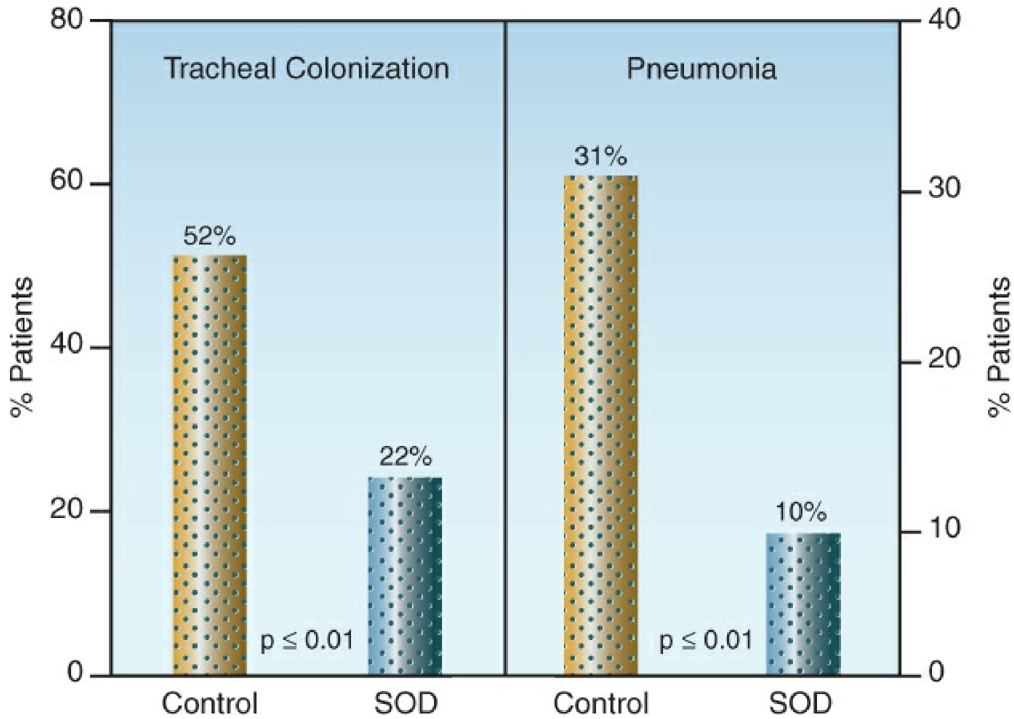

FIGURE 29.1 The effects of selective oral decontamination (SOD) on tracheal colonization and pneumonia in ventilator-dependent patients. The p value in each graph indicates a significant difference between the groups. Data from Reference 12.

The Ventilator Bundle

The Institute for Healthcare Improvement has advocated the use of “bundles” which are a collection of practices that, when used together, will improve outcomes. The bundle introduced for ventilator-dependent patients is shown in Table 29.2. Although some of the practices in this bundle do not have a direct bearing on pneumonia, its use has been shown to reduce the incidence of VAP (13), and it is sometimes called the “VAP Bundle” (14).

TABLE 29.2	The Ventilator Bundle
1. Elevate head of the bed to 30–45° above horizontal.	
2. Prophylaxis for stress ulcer bleeding.	
3. Prophylaxis for venous thromboembolism.	

4. Daily sedation holiday.
5. Daily assessment of readiness to extubate.

The Semirecumbent Position

Elevation of the head of the bed to 30–45° (semirecumbent position) is considered an important measure for preventing gastroesophageal reflux and aspiration of gastric contents into the airways. However, while some studies have validated the benefit of the semirecumbent position (15), other studies have shown no difference in the incidence of VAP with patients in the supine and semirecumbent positions (16).

COMMENT: *Since the principal source of VAP is aspiration of pathogens in mouth secretions, elevating the head of the bed could increase the risk of VAP by promoting the movement of mouth secretions into the lungs (by gravity flow).* This has never been studied, nor is it ever mentioned in discussions of the semirecumbent position to prevent VAP.

Airway Care

As mentioned in Chapter 28, tracheal tubes serve as a nidus for pathogenic colonization and biofilm formation (see Figure 28.4), and passing a suction catheter through the tubes can dislodge these biofilms and inoculate the lungs with pathogenic organisms (17). As a result of this risk, *routine suctioning of the airways is not recommended* (18). Instead, suctioning should be performed only when there is evidence of troublesome respiratory secretions.

Subglottic Aspiration

Inflation of the cuff on tracheal tubes does not prevent aspiration of mouth secretions into the lower airways, and concern about this aspiration prompted the introduction (in 1992) of specialized endotracheal tubes equipped with a suction port just above the cuff, as shown in Figure 29.2. The section port is connected to a source of continuous suction (usually not exceeding –20 cm H_2O) to clear the secretions that accumulate in the subglottic region. Clinical studies have shown *a significant reduction in the incidence of VAP when subglottic secretions are cleared using specialized endotracheal tubes* (19).

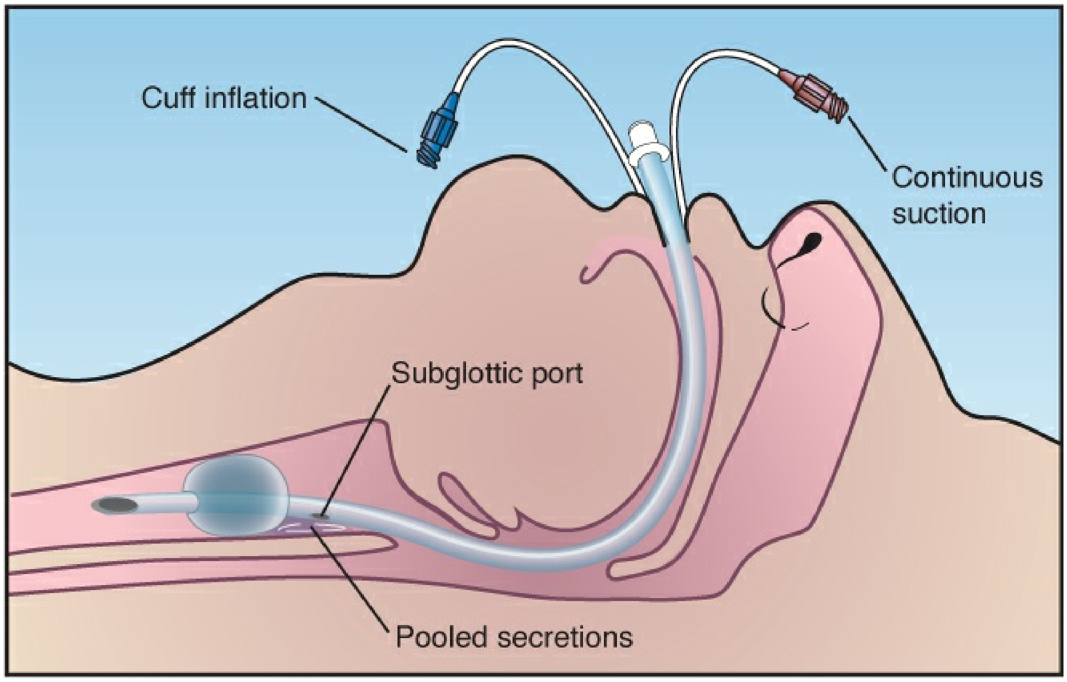

FIGURE 29.2 Endotracheal tube with a suction port placed just above the cuff to clear secretions that accumulate in the subglottic region.

CLINICAL MANIFESTATIONS

The traditional clinical criteria for the diagnosis of VAP include: (*a*) fever or hypothermia, (*b*) leukocytosis or leukopenia, (*c*), an increase in volume of respiratory secretions or a change in character of the secretions, and (*d*) a new or progressive infiltrate on the chest x-ray. The hallmark of these clinical criteria is their lack of reliability (20), as described next.

Reliability

The performance of the diagnostic criteria for VAP is shown in Table 29.3. In this case, the criteria were evaluated using cases of VAP that were diagnosed by lung biopsy (21). As indicated, none of the clinical criteria was sensitive or specific for the diagnosis of VAP, and the diagnostic odds ratio for each criterion was poor (see the legend of Table 29.3 for a description of the diagnostic odds ratio). The poor performance of these diagnostic criteria indicate that *the diagnosis of VAP is not possible using clinical criteria alone*.

TABLE 29.3 Diagnostic Value of Clinical Findings in VAP Verified by Lung Biopsy

	Sensitivity	Specificity	Diagnostic Odds Ratio†
Fever	66%	54%	2.3
Leukocytosis	64%	59%	2.6
Purulent Secretions	77%	39%	2.1

Infiltrate on Chest X-Ray	89%	26%	2.8

[†]The odds of a positive test in a patient with the disease relative to the odds of a positive test in a patient without the disease. A reliable test has an odds ratio of 10 or higher. Data from Reference 21.

Specificity of Chest Radiography

One of the major problems in the clinical diagnosis of VAP is the nonspecific nature of pulmonary infiltrates in ICU patients. This is indicated in Table 29.4, which shows a specificity of only 26% for radiographic infiltrates. *Pneumonia accounts for only one-third of the pulmonary infiltrates in ICU patients* (22,23), which means that conditions other than pneumonia are the most frequent cause of pulmonary infiltrates in ICU patients. The noninfectious causes of pulmonary infiltrates in the ICU include focal atelectasis, pulmonary edema, and the acute respiratory distress syndrome.

The images in Figure 29.3 demonstrate how portable chest x-rays can be misleading. Both chest x-rays in this figure were obtained within minutes of each other in the same patient. The x-ray on the left was taken after the patient exhaled in the supine position, and the x-ray on the right was obtained after the same patient inhaled in the upright position. Note the crowded lung markings at the right base in the image on the left, which were caused by the raised hemidiaphragm (from the supine position) and the relatively low lung volume (from the exhalation). In a patient with fever, this radiographic change could be mistaken for a basilar pneumonia.

MICROBIOLOGICAL EVALUATION

The diagnosis of VAP rests heavily on identifying a responsible pathogen, but there is no standardized method for collecting or culturing respiratory secretions. Blood cultures have limited value because *in cases of suspected VAP, organisms isolated in blood cultures are often from extrapulmonary sites of origin* (24). The following is a brief review of the methods used to collect and culture respiratory secretions.

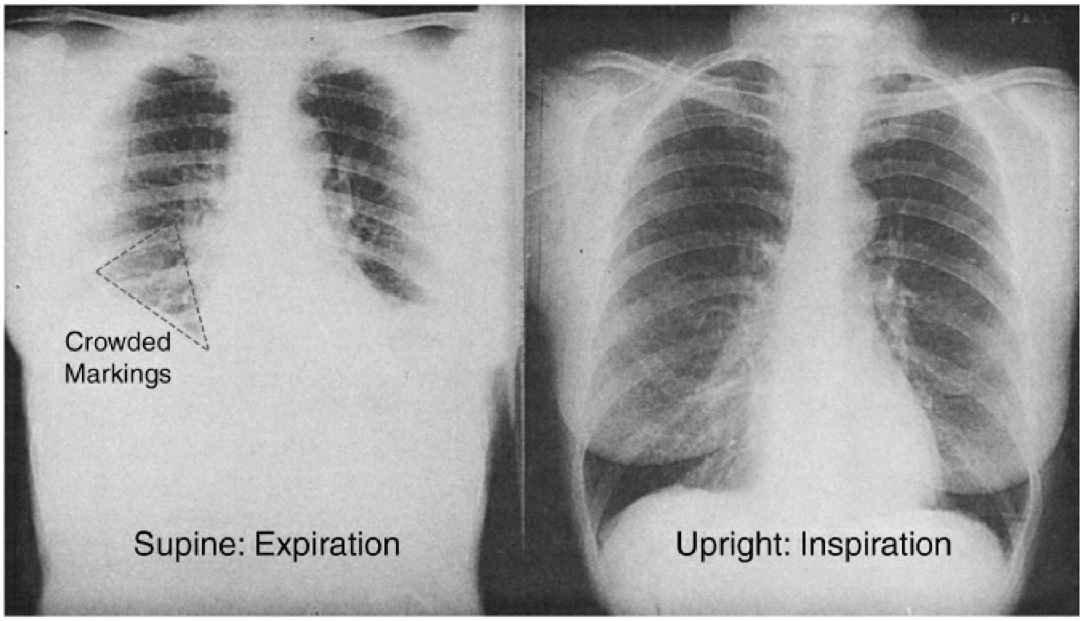

FIGURE 29.3 The effect of body position and lung volume on the appearance of a portable chest x-ray. Both images were obtained within minutes of each other in the same patient. The supine film taken after exhalation shows crowded lung markings at the right base that could be mistaken for a basilar pneumonia in a patient with a fever. Image on the left digitally enhanced.

Tracheal Aspirates

The traditional approach to suspected VAP involves aspiration of respiratory secretions through an endotracheal or tracheostomy tube, and these specimens can be contaminated with mouth secretions that are aspirated into the upper airway. A screening method to determine of the sample is appropriate for culture is described next (and should be performed routinely in the microbiology laboratory).

Microscopic Analysis

The cellular content of a tracheal aspirate is used to determine the suitability of a specimen for culture. This is explained using the sputum Gram's stain in Figure 29.4, which identifies two types of cells:

. Squamous epithelial cells, which are large, flattened cells with abundant cytoplasm and a small nucleus. These cells line the oral cavity, and *the presence of more than 10 squamous epithelial cells per low power field (×10) indicates that the specimen is contaminated with mouth secretions*, and is not an appropriate specimen for culture (25).

. Neutrophils, which are about the size of the nucleus of the epithelial cells, and have a multilobulated nucleus (which is not apparent in a low magnification view). The mere presence of neutrophils in respiratory secretions is not evidence of infection, since neutrophils can make up 20% of the cells recovered from a routine mouthwash (26). The neutrophils should be present in abundance to indicate infection; i.e., *more than 25 neutrophils per low power field (×10) is considered evidence of infection* (25). (However, this will not distinguish between tracheal bronchitis and pneumonia.)

Qualitative Cultures

The standard practice is to perform qualitative cultures on endotracheal aspirates (where the growth of organisms is reported, but there is no assessment of growth density). These cultures have a high sensitivity (usually >90%) but a very low specificity (15–40%) for identifying the culprit organism (27). This means that *for routine (qualitative) cultures of tracheal aspirates, a negative culture can be used to exclude the presence of a treatable infection, but a positive culture cannot be used as reliable evidence of the culprit organism*. The poor predictive value of positive cultures is due to contamination of tracheal aspirates with secretions from the mouth and upper airways.

FIGURE 29.4 Sputum Gram's stain under low power magnification (×10) showing the appearance of squamous epithelial cells (from the oral cavity) and neutrophils.

Quantitative Cultures

For quantitative cultures of tracheal aspirates (where growth density is reported), the threshold for the diagnosis of VAP is 10^5 colony-forming units per mL (CFU/mL) (28). This threshold has a sensitivity of 76% and a specificity of 68% for identifying the responsible pathogen (see Table 29.4) (21). Comparing these results with routine cultures (which have a sensitivity >90% and specificity ≤40%) shows that, *for cultures of tracheal aspirates, quantitative cultures are more* *reliable for identifying the responsible pathogen*.

Bronchoalveolar Lavage

Bronchoalveolar lavage (BAL) is performed by wedging a bronchoscope in a distal airway (in a lung region where the infection is suspected) and performing a lavage with sterile isotonic saline. A minimum lavage volume of 120 mL is recommended for adequate sampling (29), which is achieved by performing a series of 6 lavages using 20 mL for each lavage (only 25% or less of the volume instilled will be returned via aspiration). The first lavage is usually discarded, and the remainder of the lavage fluid is pooled and sent to the microbiology lab for microscopic analysis and quantitative culture.

Quantitative Cultures

BAL specimens are cultured quantitatively, and the threshold for the diagnosis of VAP is 10^4 CFU/mL (28). The diagnostic performance of BAL cultures is shown in Table 29.4 (21). The diagnostic odds ratio indicates that *BAL cultures are the most reliable cultures available*, but the sensitivity and specificity indicate they are far from ideal.

TABLE 29.4 Diagnostic Value of Quantitative Cultures in VAP Verified by Lung Biopsy

	Tracheal Aspirate	Protected Brush Specimen	Bronchoalveolar Lavage
Threshold (CFU/mL)	10^5	10^3	10^4
Sensitivity	76%	61%	71%
Specificity	68%	77%	80%
Diagnostic Odds Ratio†	6.6	5.1	9.6

†The odds of a positive test in a patient with the disease relative to the odds of a positive test in a patient without the disease. A reliable test has an odds ratio of 10 or higher. Data from Reference 21.

Intracellular Organisms

Inspection of BAL specimens for intracellular organisms can help in guiding initial antibiotic therapy until culture results are available. *When intracellular organisms are present in more than 3% of the cells in the lavage fluid, the likelihood of pneumonia is over 90%* (30). This is not done on a routine Gram's stain, but requires special processing and staining, and will require a specific request to the microbiology lab.

BAL Without Bronchoscopy

BAL can also be performed without the aid of bronchoscopy using a sheathed catheter like the one illustrated in Figure 29.5. This catheter is about 50 cm in length, and is inserted through an endotracheal or tracheostomy tube and advanced "blindly" until it wedges in a distal airway. The tip of the catheter has a bioabsorbable polyethylene plug that prevents contamination while the catheter is advanced. Once wedged, an inner cannula is advanced to dislodge the plug, and the BAL is performed with 10–20 mL of sterile saline.

Nonbronchoscopic BAL (also called mini-BAL because of the lower lavage volume) is a safe procedure that can be performed by respiratory therapists (31). Despite the uncertainty about the location of the catheter tip in relation to the region of suspected infection, the yield from quantitative cultures with mini-BAL is equivalent to the yield with bronchoscopic BAL (32).

Protected Specimen Brush

Aspiration of secretions through a bronchoscope produces false-positive cultures because of contamination as the bronchoscope is advanced through the upper respiratory tract (29). To eliminate this problem, a specialized brush called a *protected specimen brush* (PSB) was developed to collect uncontaminated secretions from the distal airways during bronchoscopy. Like the mini-BAL catheter, the PSB catheter has an inner cannula and a bioabsorbable plug at the tip to prevent contamination during catheter advancement. When the catheter tip is placed in the desired location, the brush is advanced from the inner cannula to collect samples from the distal airways.

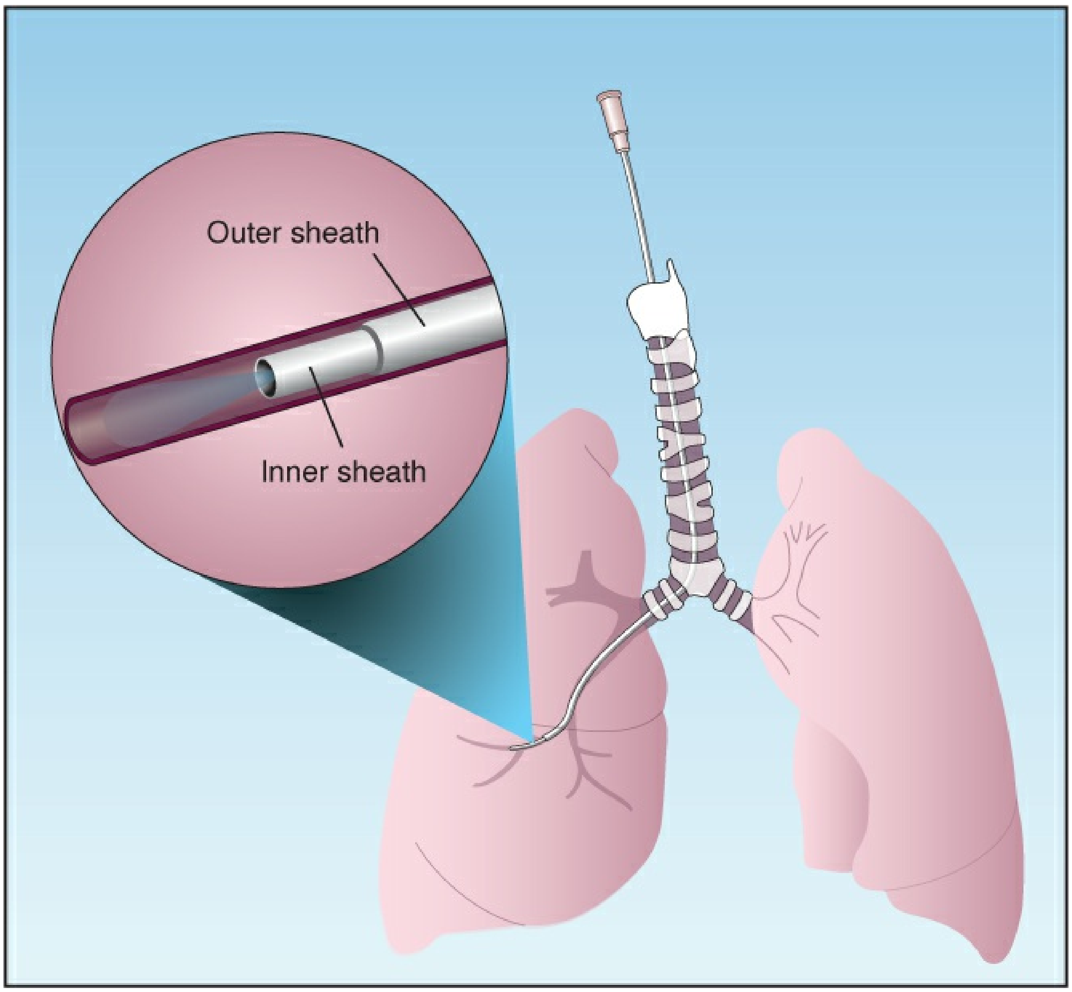

FIGURE 29.5 Protected catheter for performing bronchoalveolar lavage without the aid of bronchoscopy. See text for explanation.

Quantitative Cultures

PSB cultures are processed quantitatively, and the threshold for a positive culture is 10^3 CFU/mL (21). The diagnostic performance of PSB cultures is shown in Table 29.4 (21); as indicated by the diagnostic odds ratio, PSB cultures are less reliable than BAL cultures, but are more specific than quantitative cultures of tracheal aspirates.

Which Method is Preferred?

The clinical practice guidelines differ about the appropriate culture method: i.e.,

- The international guidelines (1) recommend quantitative cultures with BAL, mini-BAL, or PSB over cultures of tracheal aspirates, which optimizes accuracy, and reduces the number of false-positive cultures that result in overuse of antibiotics. These cultures, however, must be obtained prior to the start of empiric antibiotic therapy.
- The U.S.-based guidelines (2) recommend routine tracheal cultures over all other culture methods, based on evidence that the more invasive procedures (BAL or PSB) do not improve clinical outcomes. Once again, these cultures should be obtained prior to the start of empiric antibiotic therapy.

Take your pick, although it seems that the culture methods with the best performance characteristics (i.e., BAL cultures) should be preferred, and using mini-BAL would avoid the extra cost of bronchoscopy.

PARAPNEUMONIC EFFUSIONS

Pleural effusions are present in 20–40% of pneumonias in hospitalized patients (33). These *parapneumonic effusions* can be evident on chest x-ray, although ultrasound is recommended to define the location and size of the effusion (34), especially if thoracentesis is performed.

Classification

Parapneumonic effusions can be placed in 4 categories of increasing severity. These are outlined in Table 29.5 (33).

TABLE 29.5 Classification of Parapneumonic Effusions

	Character of Effusion	Pleural Fluid Analysis	Chest Tube
Category 1	<10 mm in thickness	Thoracentesis not necessary	No
Category 2	>10 mm thick but <50% of hemithorax, free-flowing.	pH >7.20, Glucose >60 mg/dL, negative Gram's stain & culture.	No
Category 3	Loculated, or fills >50% of hemithorax.	pH <7.20, Glucose <60 mg/dL, positive Gram's stain or culture.	Yes
Category 4	Purulent	Same as Category 3	Yes

From Reference 33.

Category 1

Category 1 effusions are small (<10 mm on decubitus films, CT images, or ultrasound), and are free-flowing. No thoracentesis is required, unless the patient's clinical condition deteriorates and the effusion increases in size (see next).

Category 2

Category 2 effusions are small-to-moderate in size (>10 mm thickness, but fill less than one-half the hemithorax) and are free-flowing. Thoracentesis is advised, and as much fluid as possible should be removed. (*Note:* Some advise measuring the pleural pressure with a manometer during pleural drainage, and stopping the drainage if the pleural pressure reaches –20 cm H_2O, to prevent negative-pressure pulmonary edema) (34). Pleural fluid analysis in category 2 effusions shows no evidence of infection: i.e., the pleural fluid pH is >7.20, the glucose is >60 mg/dL, and the Gram's stain and culture are negative. No other intervention is needed for these effusions.

Category 3

Category 3 effusions (are also called "complicated effusions") are either loculated, or they fill more than one-half of the hemithorax, and pleural fluid analysis reveals evidence of infection: i.e., the pH is <7.20, the glucose is <60 mg/dL, and the Gram's stain and/or culture is positive. Pleural drainage is required for these effusions, using small-bore (8.5–14 French) chest tubes.

Category 4

Category 4 effusions are empyemas; i.e., the pleural fluid is grossly purulent. Immediate pleural drainage is essential. Some prefer large (28–36 French) chest tubes for draining pus, but small bore (8.5–14 French) tubes achieve drainage that is equivalent to the larger tubes (35).

Antibiotics

Pleural drainage of infected parapneumonic effusions should be combined with empiric antibiotic therapy. For parapneumonic effusions in hospitalized patients, empiric coverage should include methicillin-resistant *Staphylococcus aureus*, and gram-negative enteric organisms, including *Pseudomonas aeruginosa* (36). A suitable regimen would be vancomycin plus cefepime or piperacillin/tazobactam.

Ineffective Drainage

Loculated effusions can be difficult to drain. If drainage is inadequate with the initial chest tube, an additional tube can be placed (preferably under CT guidance). Further problems with drainage should prompt a surgical consult for *video-assisted thoracoscopic surgery* (VATS), which can disrupt the pleural adhesions without a thoracotomy. The intrapleural administration of fibrinolytic agents is generally not advised for complicated parapneumonic effusions and empyema (36,37).

ANTIMICROBIAL THERAPY

Prompt administration of empiric antibiotics is recommended for all cases of suspected VAP.

The antibiotic regimen is determined by the risk factors described next.

Risk Factors

The risk factors for an unfavorable outcome include all of the following (1,2): (*a*) late-onset VAP, (*b*) septic shock, (*c*) prior infection with methicillin-resistant *Staphylococcus. aureus* (MRSA) or a multidrug-resistant organism (i.e., resistant to at least two different classes of antibiotics), (*d*) a hospital microbiogram in which 25% of the isolates are resistant organisms, and (*e*) antibiotic exposure in the past 90 days.

Empiric Regimens

The recommended empiric regimens for low-risk and high-risk patients are outlined in Table 29.6 (1,2), and are briefly summarized below.

Low-Risk Regimen

Early-onset cases of VAP without septic shock or any of the other risk factors just mentioned can be treated with a single agent that covers methicillin-sensitive *Staphylococcus aureus* (MSSA), gram-negative enteric organisms, and *Pseudomonas aeruginosa* (1,2). Candidates include cefepime, levofloxacin, and piperacillin/tazobactam.

High-Risk Regimen

The high-risk empiric regimen includes coverage for MRSA (with vancomycin or linezolid) and antipseudomonal coverage with a β-lactam antibiotic (cefepime, levofloxacin, or piperacillin/tazobactam). If there is a high index of suspicion for a multidrug-resistant organism, additional gram-negative and antipseudomonal coverage is recommended using a non-β-lactam agent (a fluoroquinolone or an aminoglycoside) (1).

There is a tendency to continue empiric antibiotics despite negative culture results if patients are improving, but this is not justified, and antibiotics should be discontinued if all cultures are unrevealing.

TABLE 29.6 Empiric Antibiotic Regimens for Suspected VAP

Low Risk Patients	High Risk Patients
A. MSSA and gram-negative (including antipseudomonal) coverage with any of the following: 1. Cefepime 2. Levofloxacin 3. Piperacillin/Tazobactam	A. MRSA coverage with: 1. Vancomycin or 2. Linezolid B. Gram-negative and antipseudomonal coverage with a β-lactam agent: 1. Piperacillin/Tazobactam or 2. Cefepime or 3. Meropenem C. Gram-negative and antipseudomonal coverage with a non-β-lactam agent:[§] 1. Levofloxacin or 2. An aminoglycoside

§Include regimen C only if there is a high risk of infection with multidrug-resistant organisms.

MSSA = methicillin-sensitive *S. aureus*. Recommendations from References 1 and 2.

Treatment of Documented Pneumonia

Antibiotic therapy of documented VAP will be dictated by the sensitivities of the isolated pathogen(s). The recommended duration of antibiotic therapy is 7 days (1,2).

A FINAL WORD

The Mortality Risk with VAP

As mentioned early in this chapter, a number of studies have shown that VAP has little or no impact on mortality rates (6–8). This observation suggests one of the following scenarios:

- We're really good at treating VAPs.
- VAPs are not life-threatening infections.
- VAPs are overdiagnosed.

Recalling the lack of specificity in the diagnostic criteria for VAP (see Tables 29.3 and 29.4), it seems that the third choice is the most appropriate.

References

Clinical Practice Guidelines

1. Torres A, Niederman MS, Chastre J, et al. International ERS/ESICM/ESCMID/ALAT guidelines for the management of hospital-acquired pneumonia and ventilator-associated pneumonia. Eur Respir J 2017; 50:1700582.
2. Kalil AC, Metersky ML, Klompas M, et al. Management of adults with hospital acquired and ventilator-associated pneumonia: 2016 Clinical Practice Guidelines by the Infectious Diseases Society of America and the American Thoracic Society. Clin Infect Dis 2016; 63:e61–e111.

General Information

3. Koulenti D, Tsigou E, Rello J. Nosocomial pneumonia in 27 ICUs in Europe: perspectives from the EU-VAP/CAP study. Eur J Clin Microbiol Infect Dis 2017; 36:1999–2006.
4. Hong HL, Hong SB, Ko GB, et al. Viral infection is not uncommon in adult patients with severe hospital-acquired pneumonia. PLoS One 2014; 9:e95865.
5. Papazian L, Klompas M, Luyt C-L. Ventilator-associated pneumonia in adults: a narrative review. Intensive Care Med 2020; 46:888–906.
6. Bregeon F, Cias V, Carret V, et al. Is ventilator-associated pneumonia an independent risk factor for death? Anesthesiology 2001; 94:554–560.
7. Bekaert M, Timsit JF, Vansteelandt S, et al. Attributable mortality of ventilator-associated pneumonia: a reappraisal using causal analysis. Am J Respir Crit Care Med 2011; 184:1133–1139.
8. Forel JM, Voillet F, Pulina D, et al. Ventilator-associated pneumonia and ICU mortality in severe ARDS patients ventilated according to a lung-protective strategy. Crit Care 2012; 16:R65

Preventive Measures

9. Estes RJ, Meduri GU. The pathogenesis of ventilator associated pneumonia: I. Mechanisms of bacterial transcolonization and airway inoculation. Intensive Care Med 1995; 21:365–383.
10. Price R, MacLennan G, Glen J. Selective digestive or oropharyngeal decontamination and topical oropharyngeal chlorhexidine for prevention of death in general intensive care: systematic review and network meta-analysis. BMJ 2014; 348:g219.
11. Bellissimo-Rodrigues WT, Menengueti MG, de Macedo LD, et al. Oral mucositis as a pathway for fatal outcome among

critically ill patients exposed to chlorhexidine: post hoc analysis of a randomized clinical trial. Crit Care 2019; 23:382.
12. Bergmans C, Bonten M, Gaillard C, et al. Prevention of ventilator-associated pneumonia by oral decontamination. Am J Respir Crit Care Med 2001; 164:382–388.
13. Al-Tawfiq JA, Abed MS. Decreasing ventilator-associated pneumonia in adult intensive care units using the Institute for Healthcare Improvement bundle. Am J Infect Control 2010; 38:552–556.
14. Wip C, Napolitano L. Bundles to prevent ventilator-associated pneumonia: how valuable are they? Curr Opin Infect Dis 2009; 22:159–166.
15. Drakulovic MB, Torres A, Bauer TT, et al. Supine body position as a risk factor for nosocomial pneumonia in mechanically ventilated patients: a randomized trial. Lancet 1999; 354:1851–1858.
16. van Nieuwenhoven CA, Vandenbroucke-Grauls C, van Tiel FH, et al. Feasibility and effects of the semirecumbent position to prevent ventilator-associated pneumonia: a randomized study. Crit Care Med 2006; 34:396–402.
17. Gil-Perontin S, Ramirez P, Marti V, et al. Implications of endotracheal tube biofilm in ventilator associated pneumonia response: a state of concept. Crit Care 2012; 16:R93.
18. Blakeman TC, Scott JB, Yoder MA, et al. AARC clinical practice guidelines: artificial airway suctioning. Respir Care 2022; 67:258–271.
19. Muscedere J, Rewa O, McKechnie K, et al. Subglottic secretion drainage for the prevention of ventilator-associated pneumonia: a systematic review and meta-analysis. Crit Care Med 2011; 39:1985–1991.

Clinical Manifestations

20. Wunderink RG. Clinical criteria in the diagnosis of ventilator-associated pneumonia. Chest 2000; 117: 191S–194S.
21. Fernando SM, Tran A, Cheng W, et al. Diagnosis of ventilator-associated pneumonia in critically ill adult patients – a systematic review and meta-analysis. Intensive Care Med 2020; 46:1170–1179.
22. Louthan FB, Meduri GU. Differential diagnosis of fever and pulmonary densities in mechanically ventilated patients. Semin Resp Infect 1996; 11:77–95.
23. Singh N, Falestiny MN, Rogers P, et al. Pulmonary infiltrates in the surgical ICU: prospective assessment of predictors of etiology and mortality. Chest 1998; 114:1129–1136.

Microbiological Evaluation

24. Luna CM, Videla A, Mattera J, et al. Blood cultures have limited value in predicting severity of illness and as a diagnostic tool in ventilator-associated pneumonia. Chest 1999; 116:1075–1084.
25. Wong LK, Barry AL, Horgan S. Comparison of six different criteria for judging the acceptability of sputum specimens. J Clin Microbiol 1982; 16:627–631.
26. Rankin JA, Marcy T, Rochester CL, et al. Human airway macrophages. Am Rev Respir Dis 1992; 145:928–933.
27. Cook D, Mandell L. Endotracheal aspiration in the diagnosis of ventilator-associated pneumonia. Chest 2000; 117:195S–197S.
28. Centers for Disease Control, National Healthcare Safety Network. Device-associated Module: Ventilator-Associated Event Protocol. January, 2013. (www.cdc.gov/nhsn)
29. Meduri GU, Chastre J. The standardization of bronchoscopic techniques for ventilator-associated pneumonia. Chest 1992; 102:557S–564S.
30. Veber B, Souweine B, Gachot B, et al. Comparison of direct examination of three types of bronchoscopy specimens used to diagnose nosocomial pneumonia. Crit Care Med 2000; 28:962–968.
31. Kollef MH, Bock KR, Richards RD, Hearns ML. The safety and diagnostic accuracy of minibronchoalveolar lavage in patients with suspected ventilator-associated pneumonia. Ann Intern Med 1995; 122:743–748.
32. Campbell CD Jr. Blinded invasive diagnostic procedures in ventilator-associated pneumonia. Chest 2000; 117:207S–211S.

Parapneumonic Effusions

33. Light RW. Parapneumonic effusions and empyema. Proc Am Thorac Soc 2006; 3:75–80.
34. Feller-Kopman D. Point: should pleural manometry be performed routinely during thoracentesis? Yes. Chest 2012; 141:844–845.
35. Rahman NM, Maskell NA, Davies CW, et al. The relationship between chest tube size and clinical outcome in pleural infection. Chest 2010; 137:536–543.
36. Shen KR, Bribiesco A, Crabtree T, et al. The American Association of Thoracic Surgery consensus guidelines for the management of empyema. J Thorac Cardiovasc Surg 2017; 153:e129–e146.
37. Aboudara M, Maldonado F. Update in the management of pleural effusions. Med Clin N Am 2019; 103:475–485.

Chapter 30

Discontinuing Mechanical Ventilation

The spirit indeed is willing, but the flesh is weak.

Matthew 26:41

Discontinuing mechanical ventilation (popularly known as *weaning* from mechanical ventilation) is often a rapid and uneventful affair, but about one of every three patients experience difficulties in the transition to unassisted breathing (1). This chapter describes the process of removing patients from mechanical ventilation, and the problems that can arise. The most recent clinical practice guidelines on this process are included in the bibliography at the end of the chapter (2–4).

VENTILATOR STRATEGIES

Attention to the following issues during mechanical ventilation can facilitate the transition to spontaneous breathing.

Patient-Triggered Breaths

The diaphragm is an involuntary muscle that contracts whenever we take a breath, and this continues as long as there is a central drive to breathe. However, mechanical ventilation can promote diaphragm weakness (5), and this *ventilator-induced diaphragm dysfunction* is particularly prominent during controlled ventilation, when the patient is not allowed to initiate a ventilator breath. This is demonstrated in Figure 30.1, which is from an animal study that compared the force of diaphragmatic contractions during spontaneous breathing, and after 3 days of assisted ventilation (where each ventilator breath was initiated by a breathing effort) or controlled ventilation (where there were no breathing efforts) (6). In this case, controlled ventilation is associated with a significant (~50%) reduction in the force of diaphragmatic contractions at each frequency tested.

Observations like those in Figure 30.1 indicate that allowing patients to trigger ventilator breaths (e.g., by avoiding neuromuscular paralysis or heavy sedation) will help to preserve the strength of the diaphragm, and should facilitate the transition from ventilatory support to spontaneous breathing. (The role of diaphragm weakness in weaning from the ventilator is described later in the chapter.)

Physiotherapy

Prolonged bed rest and physical inactivity leads to deconditioning and generalized muscle weakness, and this is considered a contributing factor in difficulties weaning from mechanical ventilation. A program of progressive physiotherapy (e.g., sitting on the edge of the bed, standing, and ambulation) has been shown to facilitate weaning from mechanical ventilation (7), and protocolized physiotherapy is recommended for all ventilator-dependent patients who are otherwise clinically stable (2).

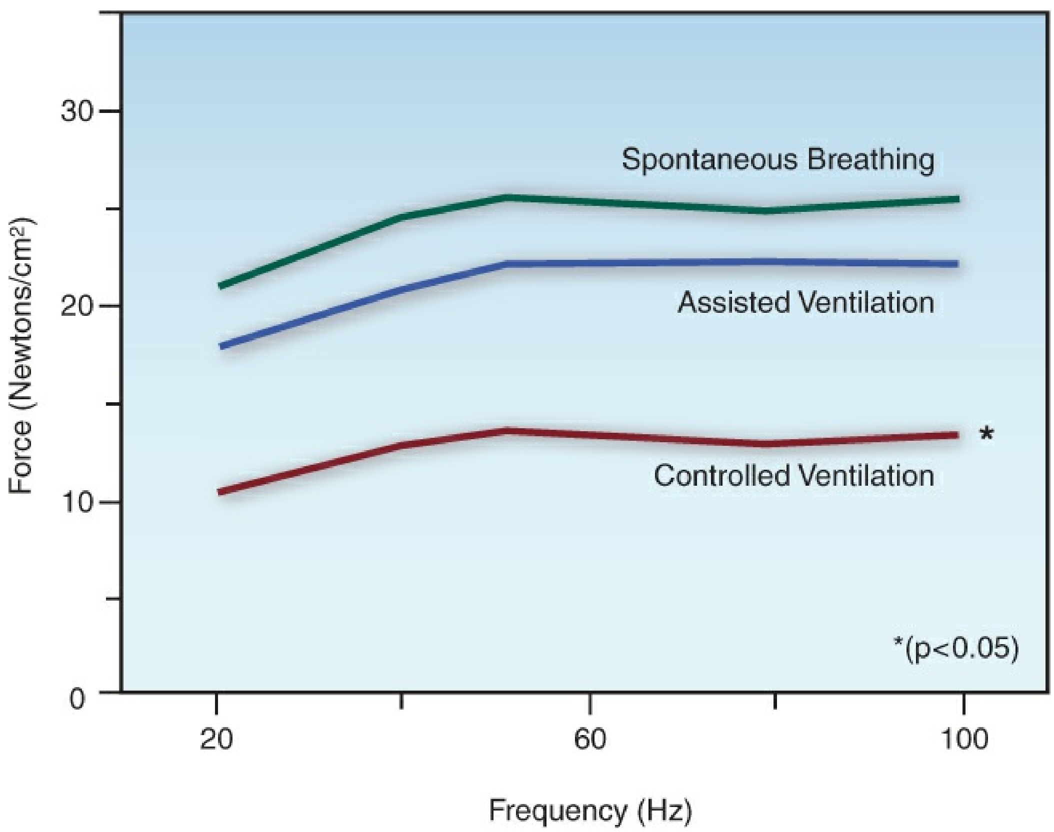

FIGURE 30.1 The force of diaphragmatic contractions at various stimulation frequences during spontaneous breathing and after 3 days of assisted ventilation (where each ventilator breath is initiated by an inspiratory effort) and controlled ventilation (where there are no spontaneous breathing efforts). Data from Reference 6.

Sedation Practices

Deep sedation (where the patient is not arousable) and sustained use of benzodiazepines (especially midazolam) will delay awakening (8), and this delays the liberation from mechanical ventilation (8). As a result, the most recent guidelines on sedation in ventilator-dependent patients (9) includes the following recommendations:

- Maintain a light level of sedation, where patients are easily aroused.
- Consider daily “sedation holidays” to prevent the accumulation of sedative agents.

. Avoid or minimize the use of benzodiazepines for sedation.

For more on sedation during mechanical ventilation, see Chapter 6.

THE SPONTANEOUS BREATHING TRIAL

Readiness

The management of ventilator-dependent patients requires constant vigilance for signs that the patient is ready for a trial of spontaneous breathing. These “readiness criteria” are listed in the left-hand column in Table 30.1. Suitable candidates should have adequate arterial oxygenation (i.e., SpO_2 ≥90%) while breathing non-toxic concentrations of oxygen (FIO_2 ≤50%) at low levels of PEEP (≤8 cm H_2O), and should have an arterial PCO_2 that is normal or at baseline levels. Patients should also be hemodynamically stable and either awake or arousable and cooperative.

TABLE 30.1 Parameters for a Spontaneous Breathing Trial

Readiness Criteria	Predictors of Success†
1. SpO_2 ≥90% with FIO_2 ≤50%	1. Tidal volume (V_T) 4–6 mL/kg (PBW)
2. PEEP ≤8 cm H_2O	2. Respiratory rate (RR) <40 /min
3. $PaCO_2$ normal or at baseline	3. RR / V_T Ratio = 60–105 bpm/L
4. Hemodynamically stable	4. Max Insp Pressure > −20 cm H_2O
5. Arousable & Cooperative	

†Measurements obtained during spontaneous breathing. PBW = predicted body weight.

When the readiness criteria are satisfied and the spontaneous breathing trial begins, the measurements listed in the right hand column of Table 30.1 can help to determine if spontaneous breathing will be successful (10). Each of these parameters can be a poor predictor of success or failure in individual patients, but when taken together, they will give you a good idea of how things will proceed. Serial changes in these measurements have more predictive value than spot measurements obtained at the onset of the spontaneous breathing trial (11).

Methods

There are two methods for a spontaneous breathing trial (SBT), as described next.

Using the Ventilator Circuit

The popular method of conducting SBTs is to leave the patient connected to the ventilator. The advantage of this method is the ability to monitor the tidal volume (V_T) and respiratory rate (RR), since rapid, shallow breathing (indicated by an increase in the RR/V_T ratio) is a common breathing pattern in patients who fail the SBT (9).

To counteract the resistance to breathing through the ventilator circuit, low levels of pressure support (5 cm H_2O) are routinely employed. (For a description of pressure support ventilation, see Chapter 26, and Figure 26.2.) However, as demonstrated in Figure 30.2, *the use of pressure* *support does not significantly reduce the work of breathing* (12), which is consistent with studies showing that the addition of pressure support does not facilitate weaning from mechanical ventilation (13).

The T-Piece Circuit

SBTs can also be conducted when the patient is disconnected from the ventilator, using the simple circuit design illustrated in Figure 30.3. A source of O_2 (usually from a wall outlet) is delivered to the patient at a high flow rate (higher than the patient's inspiratory flow rate), and this not only facilitates the inhalation of O_2, it also carries exhaled CO_2 out to the atmosphere, to prevent CO_2 rebreathing. Because this circuit employs a T-shaped adapter, this type of SBT is popularly known as a "T-piece trial".

The T-piece circuit uses high flow rates, and thus may be better suited for patients with high ventilatory demands (although this is unproven). The major disadvantage of the T-piece circuit is the inability to monitor the respiratory rate and tidal volume.

Which Method is Preferred?

There is no clinically proven advantage with any method for SBTs (13,14), which means that when a patient is ready to resume spontaneous breathing, it doesn't matter how you do it. Leaving the patient attached to the ventilator during SBTs is favored because it is easier to set up.

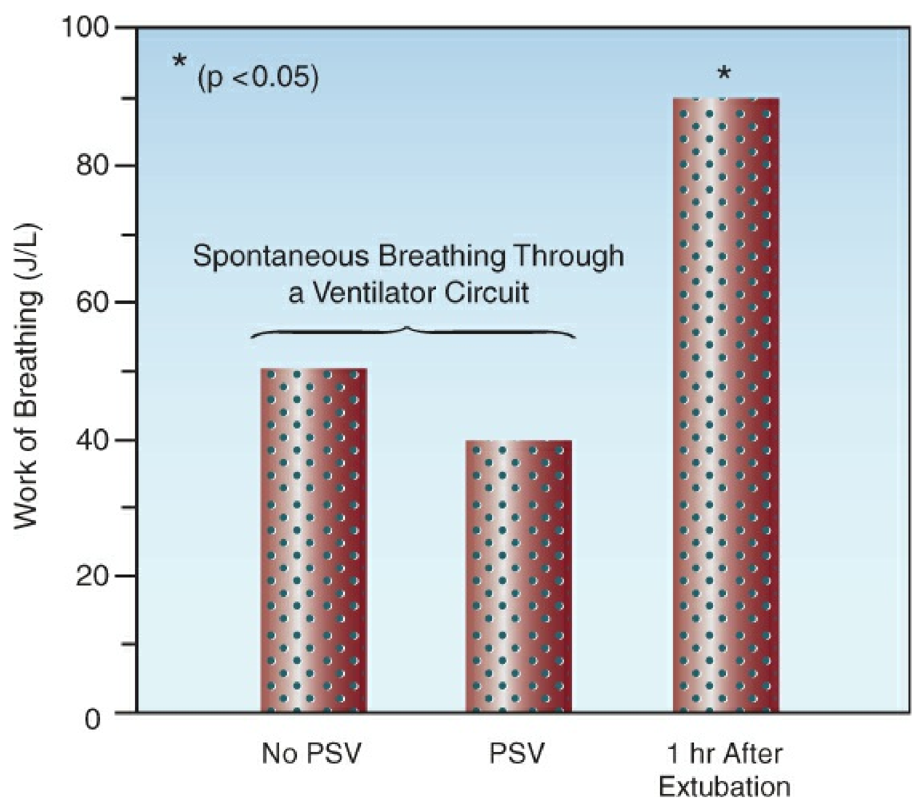

FIGURE 30.2 The work of breathing (in joules per liter) during spontaneous breathing trials conducted with and without the aid of pressure support ventilation (PSV) at 5 cm H_2O, and one hour after extubation. The asterisk indicates a significant difference at the p = 0.05 level. Data

from Reference 12.

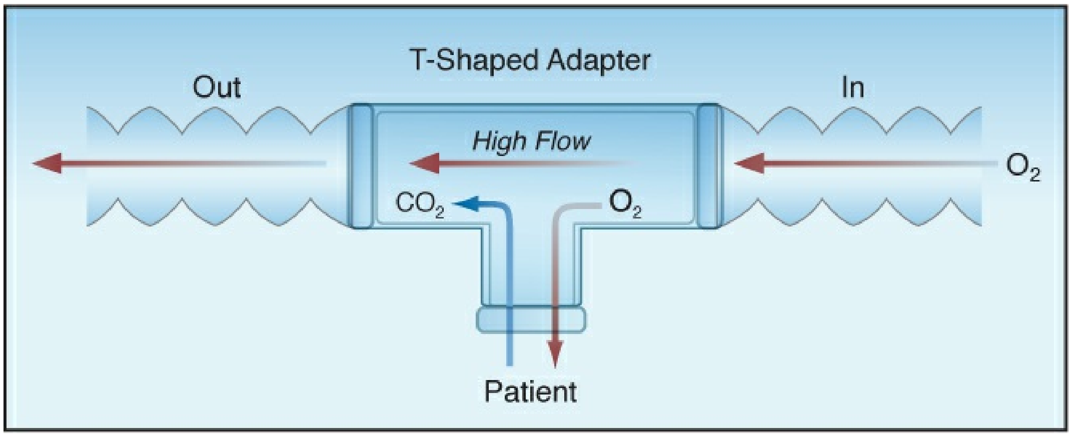

FIGURE 30.3 The T-piece circuit used for spontaneous breathing trials. The O_2 is delivered at high flow rates, which facilitates both O_2 inhalation and CO_2 removal.

Success vs. Failure

Success or failure of an SBT is judged by one or more of the following:

- Signs of respiratory distress: i.e., agitation, diaphoresis, rapid breathing, and use of accessory muscles of respiration.
- Signs of respiratory muscle weakness: e.g., paradoxical inward movement of the abdominal wall during inspiration.
- Adequacy of gas exchange in the lungs; e.g., arterial O_2 saturation, PaO_2/FIO_2 ratio, and arterial PCO_2.
- Adequacy of systemic perfusion, as assessed by the central venous O_2 saturation.

The time allotted for an SBT trial varies from less than 30 minutes to longer than 2 hours, depending on the clinical condition of the patient. A majority of patients (~80%) who tolerate SBTs for 2 hours can be permanently removed from the ventilator (10).

Rapid Breathing

Rapid breathing during SBTs is a common source of concern, but tachypnea may be a sign of anxiety rather than ventilatory failure (15). Monitoring the tidal volume can be useful in distinguishing anxiety from ventilatory failure; i.e., *anxiety produces hyperventilation*, where the respiratory rate and tidal volume are both increased, whereas ventilatory failure usually produces rapid and shallow breathing, where the respiratory rate is increased but the tidal volume is decreased. Therefore, *for the patient who develops rapid respirations during a trial of unassisted breathing, an increased tidal volume suggests anxiety as the underlying problem, while a decreased tidal volume suggests ventilatory failure*. Worsening of gas exchange may not distinguish between anxiety and ventilatory failure for the reasons described next.

Adverse Effects

Regardless of the cause, rapid breathing during SBTs can be detrimental in several ways, as summarized below.

- In patients with asthma and COPD, rapid breathing promotes hyperinflation and intrinsic PEEP, which can: (*a*) decrease the cardiac output, (*b*) increase dead space ventilation, (*c*) decrease lung compliance, and (*d*) produce diaphragm dysfunction by flattening the diaphragm.
- For patients with infiltrative lung disease (e.g., ARDS), rapid breathing reduces ventilation in diseased lung regions (where time constants for alveolar ventilation are prolonged), and this promotes alveolar collapse and hypoxemia.
- For all patients with acute respiratory failure, rapid breathing can increase whole-body O_2 consumption, which places an added burden on systemic O_2 transport.

Management

If ventilatory failure is suspected as the cause of rapid breathing, the patient should be placed back on the ventilator. If anxiety is suspected as the culprit, administration of a sedative drug should be considered. *Opiates are particularly effective in curbing the sensation of dyspnea* (16), and should be preferred in this setting. Despite the fear of opiate use in patients with COPD, opiates have been used safely for relief of dyspnea in patients with severe COPD (16).

FAILURE OF SPONTANEOUS BREATHING

A failed trial of spontaneous breathing is usually a sign that the pathologic condition requiring ventilatory support needs further improvement. However, there are other conditions that create difficulties in discontinuing mechanical ventilation, and the principal ones are described next.

Cardiac Dysfunction

Cardiac dysfunction has been identified in 40% of failed weaning trials (17). Potential sources of cardiac dysfunction in this situation include: (*a*) negative intrathoracic pressures, which increase left ventricular afterload (see Figure 26.8), (*b*) occult PEEP, which reduces venous return to the heart, (*c*) a decrease in ventricular distensibility (18) from hyperinflation and occult PEEP, and (*d*) silent myocardial ischemia (19).

The adverse effects of cardiac dysfunction include pulmonary congestion, and a decrease in the contractile strength of the diaphragm (20). The influence of cardiac output on the strength of diaphragmatic contractions is explained by the fact that the diaphragm (like the heart) maximally extracts O_2 under normal conditions, and thus is highly dependent on the cardiac output for its O_2 supply.

Monitoring

The following approaches can be used to detect cardiac dysfunction in patients who fail repeated attempts at discontinuing mechanical ventilation.

PULMONARY ARTERY CATHETER: Although rarely used, the pulmonary artery catheter would be useful for detecting weaning-induced increases in pulmonary capillary pressure and decreases in cardiac output.

CARDIAC ULTRASOUND: Cardiac ultrasound is a useful tool for detecting changes in systolic and diastolic function during failed trials of spontaneous breathing (19,21).

CENTRAL VENOUS O_2 SATURATION: A decrease in cardiac output is accompanied by a compensatory increase in peripheral O_2 extraction and a subsequent decrease in venous O_2 saturation. (See Chapter 9 for a description of the factors that influence venous O_2 saturation.) Therefore, a decrease in central venous O_2 saturation ($ScvO_2$) during a failed SBT could signal the appearance of cardiac dysfunction. The changes in *mixed* venous O_2 saturation (SvO_2) during successful and failed trials of spontaneous breathing are shown in Figure 30.4 (22). The SvO_2 decreased during the failed trials but not during the successful trials, suggesting that a decrease in cardiac output may be responsible for the failure to sustain spontaneous breathing. (*Note:* The $ScvO_2$ will mimic the behavior of the SvO_2.)

Management

There is surprisingly little information on methods for correcting the cardiac dysfunction that develops during spontaneous breathing trials, but the following interventions should be considered:

CPAP: Patients who develop systolic dysfunction should benefit from continuous positive airway pressure (CPAP), which promotes cardiac output by canceling the afterload-increasing effect of negative intrathoracic pressure (23,24). Since CPAP is delivered noninvasively, it will not prevent the removal of mechanical ventilation, including extubation.

DIURESIS: Patients with systolic dysfunction and an enlarged left ventricle (i.e., an increased left ventricular end-diastolic volume) or weaning-induced pulmonary edema, can benefit from diuresis (21,25). However in the absence of pulmonary edema, the judicious use of diuretics is advised for patients with concomitant right heart failure.

NITRATES: For patients who develop cardiac dysfunction associated with hypertension during an SBT, intravenous nitroglycerin (titrated to the desired blood pressure) has proved successful in removing patients from the ventilator (26).

Respiratory Muscle Weakness

Respiratory muscle weakness is always near the top of the list for potential causes of difficulty in removing ventilatory support. The following are some potential sources of respiratory muscle weakness in ventilator-dependent patients.

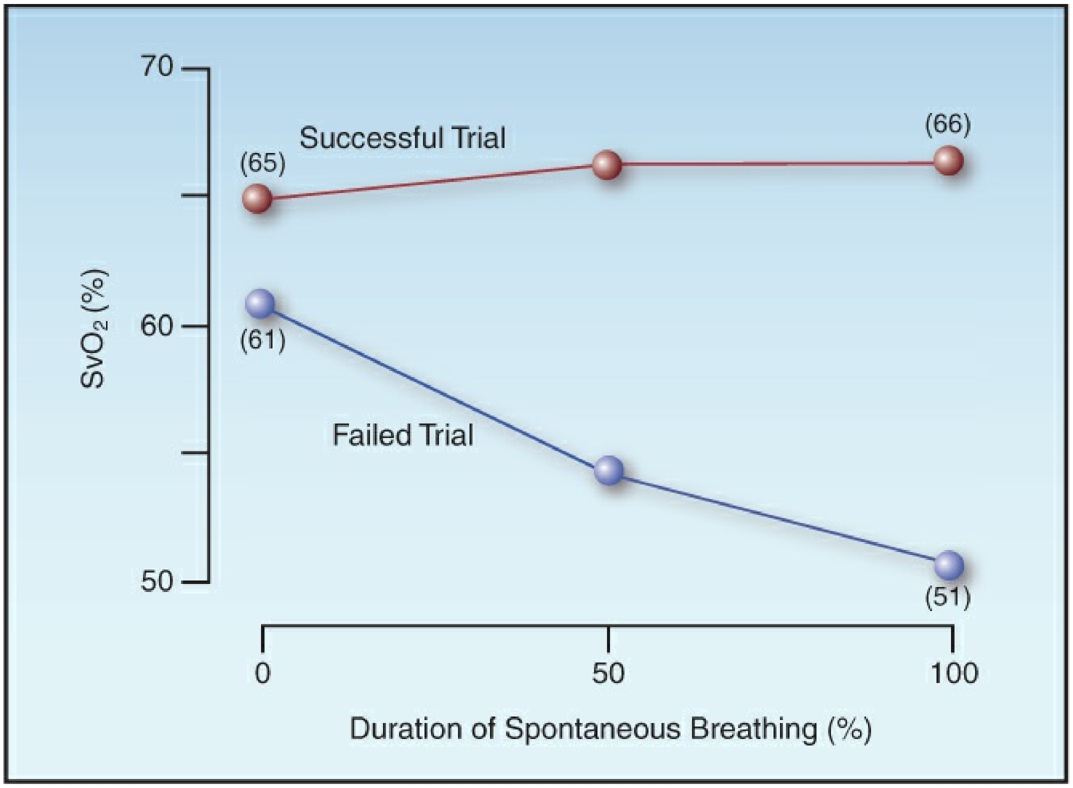

FIGURE 30.4 Mixed venous O_2 saturation (SvO_2) during successful and failed trials of spontaneous breathing. Data from Reference 22.

Potential Sources

MECHANICAL VENTILATION: As mentioned earlier (and shown in Figure 30.1), mechanical ventilation can cause diaphragm weakness when patients are not allowed to trigger ventilator breaths.

CRITICAL ILLNESS NEUROMYOPATHY: The condition known as *critical illness neuromyopathy* is an inflammatory injury to peripheral nerves and/or skeletal muscle that appears in critically ill patients, and is often recognized only when patients fail to wean from mechanical ventilation (27). This condition is described in more detail in Chapter 46. There is no effective treatment.

ELECTROLYTE DEPLETION: Magnesium and phosphorous depletion can promote respiratory muscle weakness (28,29), but the clinical relevance of this effect is unproven. Nevertheless, deficiencies in these electrolytes should be corrected in patients who fail attempts to discontinue mechanical ventilation.

Monitoring

In the supine position, diaphragm weakness can be detected by inward movement of the abdomen during inspiration, but there are few quantitative measures of diaphragm strength that are clinically applicable. (Measures such as transdiaphragmatic pressure and response to phrenic nerve stimulation are not routinely available, and are not described here.)

MAXIMUM INSPIRATORY PRESSURE: The standard clinical measure of respiratory muscle strength is the maximum inspiratory pressure (PI_{max}), which is the negative pressure that is generated by a maximum inspiratory effort against a closed airway. The normal values of PI_{max} vary widely, but mean values of –120 cm H_2O and –84 cm H_2O have been reported for adult men and women, respectively (30). Ventilation at rest is threatened when the PI_{max} drops to –20 or –30 cm H_2O, which is at the threshold for predicting the success of spontaneous breathing trials (see Table 30.1). Patients with acute respiratory failure often have difficulty performing the maneuver needed to measure PI_{max}, which limits its value is assessing diaphragm weakness in the ICU.

ULTRASOUND: Ultrasound measures of diaphragm thickness and diaphragm excursion have been proposed as measures of diaphragm dysfunction in ICU patients (31), but this methodology is limited by the lack of standardized reference measurements and the paucity of sonographers skilled in diaphragm imaging. Furthermore, the impact of this imaging on weaning from the ventilator is unclear.

Management

The management of respiratory muscle weakness is limited to physiotherapy and general supportive care (nutrition, etc.).

EXTUBATION

Once there is evidence that mechanical ventilation is no longer necessary, the next step is to remove the artificial airway. This section focuses on removing endotracheal tubes, although some of the principles also apply to removing tracheostomy tubes. (The removal of tracheostomy tubes is a more gradual process, and often occurs after patients leave the ICU.) Extubation should never be performed to reduce the work of breathing (in patients who are experiencing some respiratory distress during weaning), because the work of breathing can actually *increase* after extubation, as demonstrated in Figure 30.2. (This can be the result of an increased respiratory rate or breathing through a narrowed glottis, but it occurs in patients who tolerate extubation, so it is not necessarily a cause for concern.)

The considerations that must be addressed prior to extubation include: (*a*) the patient’s ability to clear secretions from the airways, and (*b*) the risk of symptomatic laryngeal edema following extubation.

Airway Protective Reflexes

The ability to protect the airway from aspirated secretions is determined by the strength of the gag and cough reflexes. Cough strength can be assessed by holding a piece of paper 1–2 cm from the end of the endotracheal tube and asking the patient to cough. If wetness appears on the paper, the cough strength is considered adequate (32). Diminished strength or even absence of cough or gag reflexes will not necessarily prevent extubation, but will identify patients who at risk of aspiration. These patients deserve special attention aimed at swallowing function post-extubation.

Laryngeal Edema

Trauma to the larynx from the insertion and presence of endotracheal tubes can lead to laryngeal edema and upper airway obstruction after extubation. The reported incidence of laryngeal edema that is severe enough to warrant reintubation is 1–10% (33), and contributing factors include difficult and prolonged intubation, large tube size, and unplanned self-extubation.

The Cuff-Leak Test

The cuff-leak test is designed to determine the risk of upper airway obstruction from laryngeal edema after the endotracheal tube is removed. The test is performed while the patient is receiving ventilator breaths. The cuff on the endotracheal tube is deflated, which normally causes air to leak out of the lungs, and the volume of the air leak can be measured as the difference between the inhaled and exhaled volumes. (This test is best performed during volume-controlled ventilation, where the inflation volume is pre-selected.) The absence of an air leak when the cuff is deflated is then considered evidence of an upper airway obstruction from laryngeal edema.

PROBLEMS: The cuff leak test was developed for extubating children with croup (34), and the value of the test in adults has been debated for years. One of the problems is defining a negative test (i.e., evidence that upper airway obstruction is unlikely), since some studies use the presence of an auditory leak as a negative test, while other studies use a specified volume of gas that is leaked (typically 100–110 mL) as a negative test. The other problem has been the limited predictive value of positive and negative tests for identifying troublesome laryngeal edema post-extubation (32,35).

RECOMMENDATION: The cuff test should be reserved for patients that are high-risk for post-extubation laryngeal edema (e.g., those with prolonged or difficult intubations). Based on a recent study in this patient population (35), the following recommendations are reasonable:

- A leak volume >110 mL eliminates the risk of post-extubation laryngeal edema with about 95% certainty, and nothing further is needed prior to extubation.
- A leak volume <110 mL increases the risk of post-extubation laryngeal edema 7-fold, and these patients should receive pretreatment with steroids prior to extubation. They should also be placed on noninvasive ventilation immediately after extubation.

Pretreatment with Steroids

There is evidence that pretreatment with intravenous corticosteroids reduces the risk of post-extubation laryngeal edema (36,37). This is demonstrated in Figure 30.5, which is from a large, multicenter study showing that intravenous methylprednisolone (20 mg every 4 hours, with the first dose given 12 hours prior to a planned extubation) was associated with marked reductions in post-extubation laryngeal edema and in the reintubation rate (36). *Steroid pretreatment must begin hours before the extubation*, since the effect is not present if steroids are administered close to the time of extubation (38).

Noninvasive Ventilation

Noninvasive ventilation (which is described in Chapter 26) is effective in reducing the rate of reintubation when used immediately after extubation, in patients with a high risk of laryngeal edema (39). However, similar success has not been demonstrated when post-extubation respiratory failure is established (40). Therefore, noninvasive ventilation should be used immediately after extubation in high-risk patients, and is not recommended once respiratory failure is established.

Post-extubation Stridor

The characteristic sign of upper airway obstruction from laryngeal edema is noisy breathing, also called *stridorous breathing*, or simply *stridor*, which is first apparent when the obstruction reaches about 50% (33). The sounds may be high-pitched and wheezy, or low-pitched and harsh, but they are always audible without a stethoscope, and they are *always most apparent during inspiration*. This inspiratory prominence is due to the extrathoracic location of laryngeal obstruction: i.e., negative intrathoracic pressures generated during a spontaneous inspiration are transmitted to the upper airways outside the thorax, and this results in a narrowing of the extrathoracic airways (including the larynx and pharynx) during the inspiratory phase of breathing. (In contrast, the intrathoracic airways narrow during expiration.)

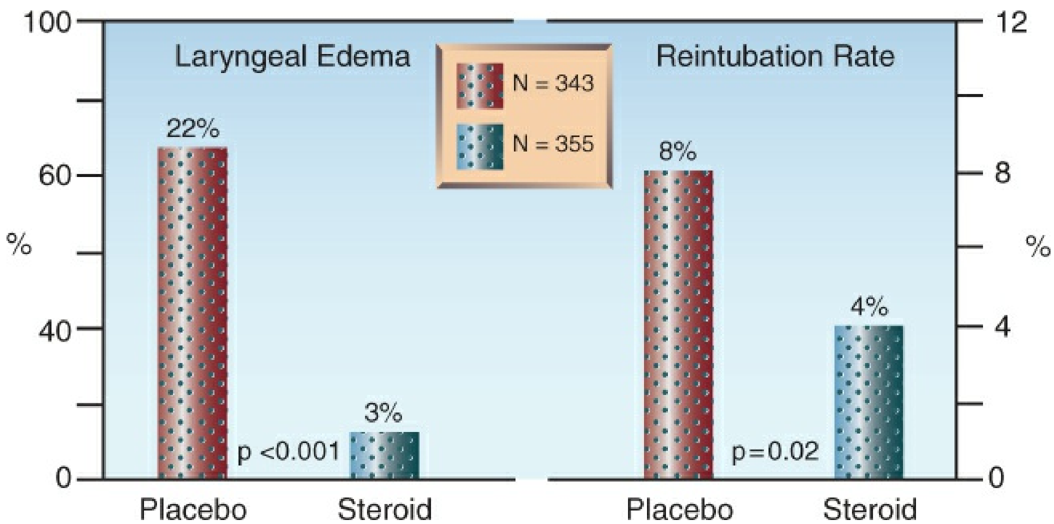

FIGURE 30.5 Results of a large, multicenter study showing that pretreatment with methylprednisolone (20 mg IV every 4 hrs, starting 12 hours prior to extubation) was associated with a 7-fold drop in the incidence of post-extubation laryngeal edema, and a 50% decline in the reintubation rate. Data from Reference 36.

Management

Post-extubation stridor is apparent within 30 minutes of extubation in most cases (32), but delays of up to 2 hours can occur (personal observation). Reintubation is not always necessary, but *any signs of impending respiratory failure (O_2 desaturation, rapid breathing) should prompt immediate reintubation*, because delays to reintubation can make it difficult to re-establish a stable airway. Furthermore, *there are no proven treatments for post-extubation laryngeal edema* (33).

Popular but unproven treatments for post-extubation laryngeal edema include nebulized epinephrine and intravenous steroids. Since there is no evidence that either of these interventions is effective, it is not possible to recommend an effective dosage.

A FINAL WORD

Avoiding Delays

One of the most important aspects of discontinuing mechanical ventilation is to avoid unnecessary delays, and the following measures should help in this regard.

- Keep sedation as light as possible.
- Start physiotherapy as soon as the patient is clinically stable.
- Be attentive to the patient's nutritional status.
- Be vigilant for signs that the patient is ready for a spontaneous breathing trial, and extubate the patient in a timely fashion if the trial is successful.
- Don't forget the possible role of the heart in failed trials of spontaneous breathing.
- If concerned about post-extubation laryngeal edema, pretreat with steroids prior to extubation, and extubate the patient to noninvasive positive pressure breathing.

These measures should help to free patients from mechanical ventilation, and end the misfortune of being tethered to a machine.

References

1. Pham T, Heunks L, Bellani G, et al. Weaning from mechanical ventilation in intensive care units across 50 countries (WEAN SAFE): a multicenter, prospective, observational cohort study. Lancet Respir Med 2023; 11:465–476.

Clinical Practice Guidelines

2. Girard TD, Alhazzani W, Kress JP, et al. An official American Thoracic Society/American College of Physicians clinical practice guideline: Liberation from mechanical ventilation in critically ill adults. Am J Respir Crit Care Med 2017; 195:120–133.
3. Schmidt GA, Girard TD, Kress JP, et al. Official executive summary of an American Thoracic Society/ American College of Physicians clinical practice guideline: Liberation from mechanical ventilation in critically ill adults. Am J Respir Crit Care Med 2017; 195:115–119.
4. Robert KJ, Goodfellow LT, Battey-Muse CM, et al. AARC Clinical Practice Guideline: Spontaneous breathing trials for liberation from adult mechanical ventilation. Respir Care. March, 2024 (epub ahead of print).

Preliminary Concerns

5. Wu H, Chasteen B. Rapid review of ventilator-induced diaphragm dysfunction. Respir Med 2024; 223:107541.
6. Sassoon CSH, Zhu E, Caiozzo VJ. Assist-control mechanical ventilation attenuates ventilator-induced diaphragm dysfunction. Am J Respir Crit Care Med 2004; 170:626–632.
7. Schreiber AF, Ceriana P, Ambrosino N, et al. Physiotherapy and weaning from prolonged mechanical ventilation. Respir Care 2019; 64:17–25.
8. Pohlman AS, Simpson KP, Hall JB. Continuous intravenous infusions of lorazepam versus midazolam for sedation during mechanical ventilatory support: a prospective, randomized study. Crit Care Med 1994; 22:1241–1247.
9. Devlin JW, Skrobik Y, Gélinas C, et al. Clinical practice guidelines for the prevention and management of pain, agitation/sedation, delirium, immobility, and sleep disruption in adult patients in the ICU. Crit Care Med 2018; 48:e825–e873.

Spontaneous Breathing Trials

10. MacIntyre NR. Evidence-based assessments in the ventilator discontinuation process. Respir Care 2012; 57:1611–1618.
11. Kreiger BP, Isber J, Breitenbucher A, et al. Serial measurements of the rapid-shallow breathing index as a predictor of weaning outcome in elderly medical patients. Chest 1997; 112:1029–1034.
12. Mehta S, Nelson DL, Klinger JR, et al. Prediction of post-extubation work of breathing. Crit Care Med 2000; 28:1341–1346.
13. Thille AW, Gacouin A, Courdoy R, et al; for the REVA Research Network. Spontaneous-breathing trials with pressure-

support ventilation or a T-piece. N Engl J Med 2022; 387:1843–1854.
14. Gacouin A, Lesouhaitier M, Reizine F, et al. 1-hour t-piece spontaneous breathing trial vs 1-hour zero pressure support spontaneous breathing trial and reintubation at day 7: a non-inferiority approach. J Crit Care 2022; 67:95–99.
15. Bouley GH, Froman R, Shah H. The experience of dyspnea during weaning. Heart Lung 1992; 21:471–476.
16. Raghavan N, Webb K, Amornputtisathaporn N, O'Donnell DE. Recent advances in pharmacotherapy for dyspnea in COPD. Curr Opin Pharmacol 2011; 11:204–210.

Failure of Spontaneous Breathing

17. Grasso S, Leone A, De Michele M, et al. Use of N-terminal pro-brain natriuretic peptide to detect acute cardiac dysfunction during weaning failure in difficult-to-wean patients with chronic obstructive pulmonary disease. Crit Care Med 2007; 35:96–105.
18. Srivastava S, Chatila W, Amoateng-Adjepong Y, et al. Myocardial ischemia and weaning failure in patients with coronary artery disease: an update. Crit Care Med 1999; 27:2109–2112.
19. Papanickolaou J, Makris D, Saranteas T, et al. New insights into weaning from mechanical ventilation: left ventricular diastolic dysfunction is a key player. Intensive Care Med 2011; 37:1976–1985.
20. Nishimura Y, Maeda H, Tanaka K, et al. Respiratory muscle strength and hemodynamics in heart failure. Chest 1994; 105:355–359.
21. Goudelin M, Champy P, Amiel J-B, et al. Left ventricular overloading identified by critical care echocardiography is key in weaning-induced pulmonary edema. Intensive Care Med 2020; 46:1371–1381.
22. Jubran A, Mathru M, Dries D, Tobin MJ. Continuous recordings of mixed venous oxygen saturation during weaning from mechanical ventilation and the ramifications thereof. Am Rev Respir Crit Care Med 1998; 158:1763–1769.
23. Naughton MT, Raman MK, Hara K, et al. Effect of continuous positive airway pressure on intrathoracic and left ventricular transmural pressures in patients with congestive heart failure. Circulation 1995; 91:1725–1731.
24. Bradley TD, Holloway BM, McLaughlin PR, et al. Cardiac output response to continuous positive airway pressure in congestive heart failure. Am Rev Respir Crit Care Med 1992; 145:377–382.
25. Teboul J-L. Weaning-induced cardiac dysfunction: where are we today? Intensive Care Med 2014; 40:1069–1079.
26. Routsi C, Stanopoulos I, Zakynthinos E, et al. Nitroglycerin can facilitate weaning of difficult-to-wean chronic obstructive pulmonary disease patients: a prospective interventional non-randomized study. Crit Care 2010; 14:R204.
27. Vanhorebeek I, Latronico N, Van den Berghe G. ICU-acquired weakness. Intensive Care Med 2020; 46:637–653.
28. Benotti PN, Bistrian B. Metabolic and nutritional aspects of weaning from mechanical ventilation. Crit Care Med 1989; 17:181–185.
29. Malloy DW, Dhingra S, Solren F, et al. Hypomagnesemia and respiratory muscle power. Am Rev Respir Dis 1984; 129:427–431.
30. Bruschi C, Cerveri I, Zoia MC, et al. Reference values for maximum respiratory mouth pressures: A population-based study. Am Rev Respir Dis 1992; 146:790–793.
31. Zambon M, Greco M, Bocchino S, et al. Assessment of diaphragm dysfunction in the critically ill patient with ultrasound: a systematic review. Intensive Care Med 2017; 43:29–38.

Extubation

32. Khamiees M, Raju P, DeGirolamo A, et al. Predictors of extubation outcome in patients who have successfully completed a spontaneous breathing trial. Chest 2001; 120:1262–1270.
33. Pluijms WA, van Mook WNKA, Wittekamp BH, Bergmans DCJJ. Postextubation laryngeal edema and stridor resulting in respiratory failure in critically ill adult patients: updated review. Crit Care 2015; 19:295.
34. Adderley RJ, Mullins GC. When to extubate the croup patient: the "leak" test. Can J Anaesth 1987; 34:304–306.
35. Kallet RH, Matsushima A, Yoo S, Lipnick MS. Utilization of risk stratification tool and volume-based cuff leak test to assess postextubation stridor. Respir Care 2023; 68:309–319.
36. François B, Bellisant E, Gissot V, et al; for the Association des Réanimateurs du Centre-Quest (ARCO). 12-h pretreatment with methylprednisolone versus placebo for prevention of postextubation laryngeal oedema: a randomized double-blind trial. Lancet 2007; 369:1083–1089.
37. Cheng K-C, Hou C-C, Huang H-C, et al. Intravenous injection of methylprednisolone reduces the incidence of post-extubation stridor in intensive care unit patients. Crit Care Med 2006; 34:1345–1350.
38. Gaussorgues P, Boyer F, Piperno D, et al. Do corticosteroids prevent postintubation laryngeal edema? A prospective study of 276 adults. Crit Care Med 1988; 16:649–652.
39. Lin C, Yu H, Fan H, Li Z. The efficacy of noninvasive ventilation in managing postextubation respiratory failure: a meta-analysis. Heart Lung 2014; 43:99–104.
40. Esteban A, Frutos-Vivar F, Ferguson ND, et al. Noninvasive positive-pressure ventilation for respiratory failure after extubation. N Engl J Med 2004; 350:2452–2460.

Section X

ACID-BASE DISORDERS

To do common things perfectly is far better worth our endeavor than to do uncommon things respectably.

Harriet Beecher Stowe
(1864)